



Continuous prediction of knee and ankle joint angles based on sEMG and hip joint angle fusion signals by NRBO-LSTM network

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ARTICLE INFO

Keywords:

sEMG
Lower limb
Joint angle prediction
Newton-Raphson-Based Optimization
LSTM

ABSTRACT

Surface electromyography (sEMG) serves as a crucial information source for capturing human motion. Joint angle prediction based on sEMG holds significant application value in the field of rehabilitation medicine. This paper investigates continuous prediction of knee and ankle joint angles under three walking modalities (Level walking, Ramp ascent, and Stair ascent). Experimental data comprising sEMG and inertial measurement unit (IMU) signals were collected from 10 healthy subjects, with time-domain features extracted from sEMG signals and joint angles calculated from IMU data. Model performance was evaluated using the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE). Three input modalities were compared: single 7-channel sEMG signals, single 4-channel sEMG signals, and fused 4-channel sEMG with hip joint angle signals. Comparative analyses were conducted between long short-term memory networks (LSTM), MHA-LSTM (Multi-Head Attention LSTM), and NRBO-LSTM (Newton-Raphson-Based Optimization LSTM) models. Results demonstrate that when using fused signal input with the NRBO-LSTM model, the RMSE values for knee and ankle joint prediction were $4.11 \pm 0.74^\circ$ and $3.48 \pm 0.54^\circ$ (Level walking), $4.33 \pm 0.76^\circ$ and $3.6 \pm 0.78^\circ$ (Ramp ascent), $4.91 \pm 1.0^\circ$ and $4.3 \pm 0.7^\circ$ (Stair ascent), respectively. The MAE values were $3.04 \pm 0.6^\circ$ and $2.66 \pm 0.41^\circ$ (Level walking), $3.37 \pm 0.63^\circ$ and $3.29 \pm 0.61^\circ$ (Ramp ascent), $3.74 \pm 0.74^\circ$ and $3.29 \pm 0.61^\circ$ (Stair ascent), respectively. The R^2 values were $0.93 \pm 0.03/0.89 \pm 0.06$ (Level walking), $0.9 \pm 0.04/0.76 \pm 0.08$ (Ramp ascent), and $0.95 \pm 0.02/0.64 \pm 0.21$ (Stair ascent), respectively. This research provides theoretical references for exoskeleton control.

1. Introduction

In neurorehabilitation engineering, lower-limb exoskeletons are used to help individuals with neuromuscular, joint, or related functional impairments to recover. These devices facilitate motor recovery by providing physical support, guiding limb movement, and reducing biomechanical load. Assistance is delivered through the precise control of actuators at each joint, allowing real-time modulation of joint angles, angular velocities, and mechanical outputs such as force and torque [1,2]. The operational framework of a lower-limb exoskeleton consists of three primary modules: (1) a sensing system for accurately decoding of the user's motor intention; (2) a decision-making and control unit for interpreting sensory data and generating appropriate responses; and (3) actuators that execute the desired motion based on control commands

[2,3]. Accurate recognition of motor intention is essential for natural human-exoskeleton interaction, improved control precision, and enhanced rehabilitation outcomes.

In earlier studies, micro-electro-mechanical systems (MEMS) based inertial measurement units (IMUs) were employed to capture multidimensional signals, including angular velocity, linear acceleration, and magnetic field intensity, for lower-limb motion recognition [3,4]. By applying sensor fusion algorithms, these signals can be used to reconstruct the three-dimensional kinematic trajectories of the lower limbs. Surface electromyography (sEMG) records the spatiotemporal summation of motor unit action potentials (MUAPs) via electrodes placed on the skin surface. When detected over neurologically activated muscle groups, sEMG signals reflect precise motor commands generated by the central nervous system (CNS). Notably, sEMG signals emerge

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<https://doi.org/10.1016/j.bspc.2026.109978>

Received 2 September 2025; Received in revised form 28 January 2026; Accepted 20 February 2026

Available online 26 February 2026

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approximately 0 to 200 ms before to actual limb movement, exhibiting a temporal lead. Compared with IMU signals, sEMG provides a more direct and anticipatory representation of motor intention and has been widely adopted in human motion intention recognition [5–7].

Lower-limb gait intention recognition can be categorized into three main types: locomotion mode recognition, gait phase detection, and joint angle prediction. Locomotion mode recognition includes walking (level ground/ramp/stairs/variable speed), running, jumping, and other activities. [5–13]. Gait phase recognition covers stance/swing phases, or more detailed phases such as heel strike (HS), foot flat (FF), and heel off (HO) in the gait cycle [14–22]. The first two types represent discrete classification tasks, which inherently lack continuity and are therefore less suitable for enabling real-time, smooth human-robot interaction. Joint angle prediction involves analyzing the bio-mechanical coupling between muscle activity and joint movement. By directly extracting muscle-joint dynamics from sEMG signals, this method enables precise and responsive control in lower-limb assistive systems, making it a critical technique for rehabilitation and mobility assistance.

Joint angle prediction involves multiple stages, including data preprocessing, dimensionality reduction, feature extraction, and model training. The primary objective of data preprocessing is to improve signal quality by eliminating noise and artifacts, thereby ensuring the reliability of subsequent analysis. Common preprocessing techniques include band-pass filtering (e.g., 20–500 Hz), notch filtering (50 Hz or 60 Hz), signal normalization, and segmentation. Dimensionality reduction helps eliminate redundant information, reduce computational complexity, and improve the generalization performance of downstream prediction models. Commonly used methods include principal component analysis (PCA) and independent component analysis (ICA). Zhu et al. used a kernel-based enhanced PCA algorithm for dimensionality reduction to eliminate redundant information and analyze the contribution rates of principal components in lower-limb sEMG signals across different dimensional settings [23]. Wang et al. applied a deep belief network (DBN) for nonlinear dimensionality reduction of EMG features, demonstrating that the DBN approach based on neural networks offers greater adaptability compared to linear techniques such as PCA [24]. Muscle synergy analysis considers inter-muscular interactions by extracting a small set of synergy components that represent complex multi-muscle activities. As a physiological dimensionality reduction method, it more accurately reflects the underlying neuromuscular characteristics. Non-negative matrix factorization (NMF) has been employed to extract muscle synergy patterns from sEMG signals for dimensionality reduction [25,26].

Feature extraction aims to obtain signal characteristics that best represent motion patterns. These features are then used for recognition purposes. Common features include time-domain, frequency-domain, and time–frequency domain descriptors [27,28]. Time-domain features treat EMG signals as time-dependent functions and use statistical methods to compute descriptive metrics. Examples of time-domain features include root mean square, variance, mean absolute value, zero-crossing, mean absolute slope, waveform length, and slope sign changes. Frequency-domain features transform time-domain signals into the frequency domain, enabling more effective analysis of their spectral or power spectral characteristics. Compared to time-domain features, frequency-domain features are less susceptible to noise and tend to provide more stable and robust descriptors. Common frequency-domain features include mean frequency, mean peak frequency, median frequency, mean power, power spectral ratio, frequency ratio, and central frequency variance. However, the computation of frequency-domain features is relatively more complex than that of time-domain features. Time-frequency domain features based on wavelet decomposition face similar computational challenges [29]. To reduce computational complexity, time-domain features are more commonly used in practice.

Prediction methods can be divided into machine learning-based and deep learning-based approaches. The former primarily employs traditional models for prediction tasks, such as Support Vector Machines

(SVM) and Random Forests (RF) [30–32]. In contrast, deep learning-based joint angle prediction studies employ convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and their variants or hybrid architectures. These enhanced network designs have demonstrated significant improvements in prediction accuracy [26,33–36]. Owing to their exceptional capability in processing temporal data, such networks have become the predominant deep learning algorithms in joint angle estimation. For example, Yang et al. [26] achieved knee joint angle prediction during gait cycles by extracting RMS features from 3-channel sEMG signals and implementing RNN and DCNN models. Lu et al. [36] employed a convolutional LSTM (Conv-LSTM) to analyze 8-channel sEMG signals, demonstrating the superiority of time-domain features for multi-joint (hip, knee, ankle) angle prediction across seven locomotion modes, including walking and running.

Several researchers have improved the accuracy of joint angle prediction models by fusing multi-modal sensor information. For example, Han et al. [37] proposed a convolutional bidirectional LSTM (Conv-BiLSTM) for sEMG and IMU data fusion, effectively enhancing the robustness of real-time knee joint angle prediction. Xie et al. [38] improved the prediction accuracy of hip, knee, and ankle joint angles in RNN models by incorporating sEMG signals with plantar pressure data. Song et al. [39] compared the performance of two input modalities (unimodal sEMG vs. multi-modal sEMG + plantar pressure) in LSTM-based prediction models, with results demonstrating no statistically significant differences between the approaches. sEMG signals inherently detect movement intention 50–200 ms earlier than inertial IMU sensors, enabling sEMG-driven models to effectively compensate for system acquisition and prediction delays. However, in clinical applications for amputees, the limited viable sEMG acquisition sites significantly constrain the implementation of conventional multi-channel sEMG solutions. This presents several critical research challenges that demand investigation, including: whether to adopt multi-modal prediction approaches, how to optimize sEMG channel configuration under anatomical constraints, which complementary signals to select for multi-modal integration, and what prediction model architectures are most suitable.

This study investigates the prediction of knee and ankle joint angles across three gait modes. We compare the prediction performance using varying numbers of sEMG channels, different modal signals, and distinct LSTM-based models, with the goal of enhancing model suitability for patients with lower-limb muscle loss. To achieve this, we propose a novel method that fuses sEMG signals with hip joint angle data as input features and integrates the Newton-Raphson-Based Optimization (NRBO) algorithm into the LSTM architecture. The proposed method not only improves joint angle prediction accuracy but also effectively supports continuous control of lower-limb rehabilitation exoskeletons in clinical applications. The implementation pipeline of the proposed method is depicted in Fig. 1.

2. Experiment and methods

2.1. Gait data Collection

2.1.1. Participants

To ensure the accuracy and reliability of the experimental data, the experiment were conducted in a quiet and enclosed environment. This setup minimized external noise and magnetic interference, and helped maintain the participant's optimal condition. High temperatures may cause perspiration, while low temperatures may trigger shivering. The laboratory should maintain an appropriate temperature and adequate ventilation to prevent temperature-related degradation of signal quality.

Ten male participants without musculoskeletal disorders or muscle injuries participated in this study. 7-channel sEMG and 4-channel IMU signals were collected during five three patterns: level walking, ramp ascent and stair ascent. The participants had a mean age of 22.7 years, an average height of 1.741 m, and a mean weight of 72.5 kg. All

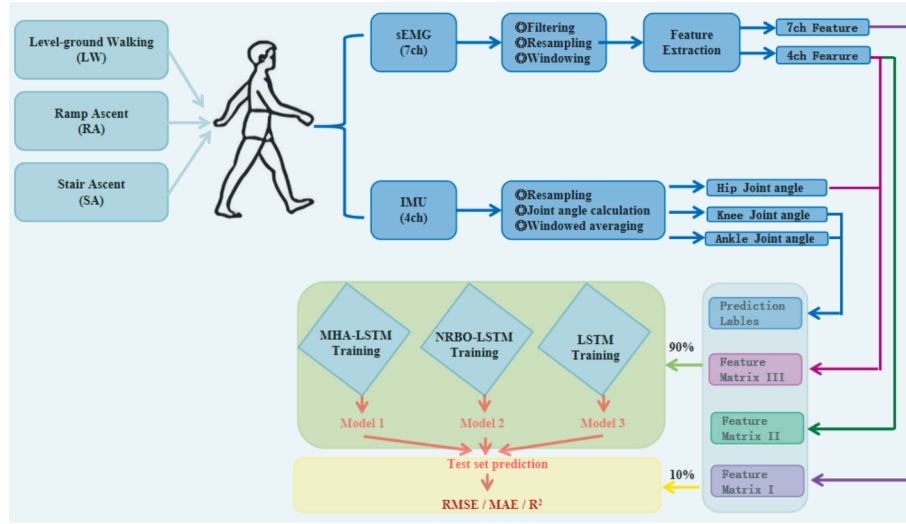


Fig. 1. Pipeline of the proposed method.

participants provided written informed consent before the study. The experimental protocol was approved by the South-Central Minzu University (Wuhan, China).

2.1.2. Data acquisition Equipment

A Delsys Trigno wireless electromyography acquisition system (Delsys Trigno System, Natick, Massachusetts, USA) was employed in this study. The system includes 16 Trigno Avanti wireless EMG sensors capable of simultaneously recording sEMG signals and 9-axis IMU data. All acquired data were transmitted to the EMGworks® software (Delsys Inc.) on a computer for real-time monitoring and storage. The sEMG signals were sampled at 1260 Hz, while the IMU data were sampled at 148 Hz. The acquisition system and sensors are illustrated in Fig. 2.

sEMG data acquisition followed standard preparation protocols to ensure signal quality and consistency: (1) The skin over the target muscle area was abraded using abrasive paste to remove surface oils and dead skin, followed by cleansing with medical alcohol to eliminate any residue. (2) Delsys wireless surface EMG electrodes were then affixed to the muscle belly using double-sided adhesive tape, with placement guided by the recommendations of the SENIAM project [40]. Electrodes were further secured using elastic sports tape. (3) After electrode placement, participants were instructed to perform a test movement according to the experimental protocol, during which sEMG signals were observed in real-time to verify correct electrode positioning.

Electrode placement avoided both the skin areas near tendons and the innervation zones of muscles, as neither location provides reliable signals. In the vicinity of tendons, muscle fibers are thinner and thus fail to generate representative activity signals. In innervation zones, the recorded signals are often unstable due to muscle contractions and elongations, leading to increased variability.

2.1.3. Selection of lower limb muscles

Lower limb movement requires the coordinated interaction of multiple muscle groups, each of which contributes differently to motion. In the shank muscles, for example, the gastrocnemius primarily facilitates ankle plantarflexion (pointing the toes downwards) and knee extension, while the tibialis anterior is responsible for ankle dorsiflexion (lifting the toes upwards) and foot inversion. The rectus femoris, located at the front of the thigh, originates from the lateral root of the thigh, the anterior inferior iliac spine, and the superior rim of the acetabulum. It wraps around the anterior patella and attaches to the tibial tuberosity on the anterior proximal tibia via the patellar tendon. Its main functions are hip flexion and knee extension. The vastus lateralis assists the rectus femoris with knee extension. The biceps femoris and semitendinosus muscles contribute to knee flexion and hip extension. These muscles work synergistically during walking on level ground, ramps, and stairs to maintain lower limb stability and flexibility. For sEMG signal acquisition, the following muscles were selected: tibialis anterior (TA), medial gastrocnemius (MGas), and lateral gastrocnemius (LGas) from the shank group; rectus femoris (RF), vastus lateralis (VL), biceps femoris (BF), and semitendinosus (ST) from the thigh group, with electrodes placed at corresponding anatomical locations. Fig. 3 illustrates the anatomical locations of the 7 sEMG electrodes.

2.1.4. Joint angle acquisition

Four sensors were placed on the participant's back, lateral thigh, lateral shank, and dorsum of the foot to acquire IMU signals, shown as Fig. 4. Joint angles were calculated from the acceleration and angular velocity data obtained from these sensors and were used as the gold standard for joint angle prediction.

The hip joint angle was defined as the sagittal plane angular



Fig. 2. The Delsys Trigno wireless sEMG acquisition system.



Fig. 3. sEMG electrode placement locations on the lower limb muscles.

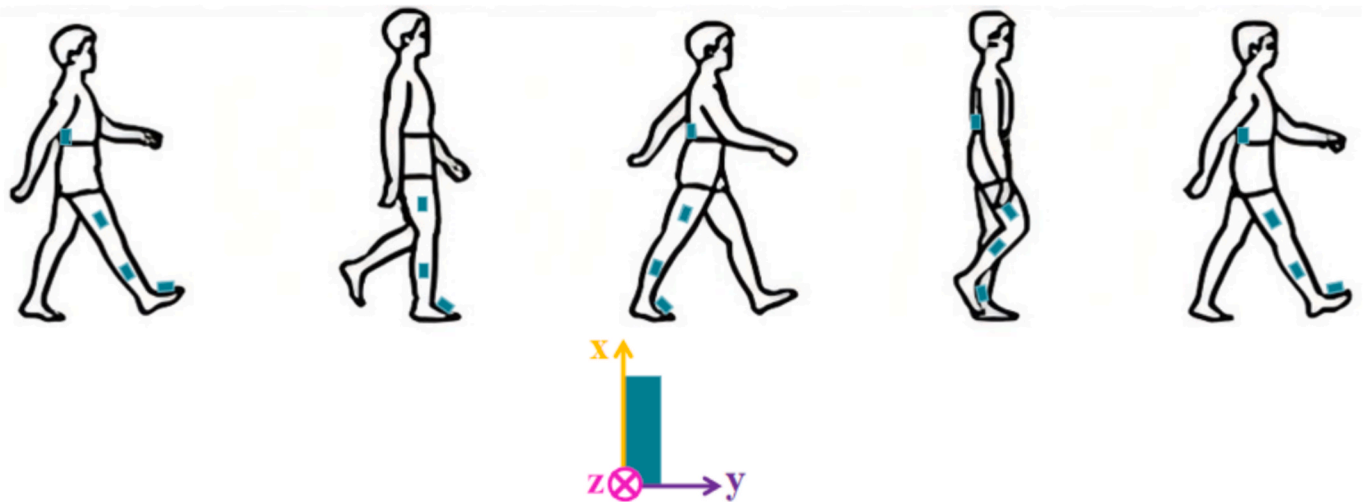


Fig. 4. IMU electrode placement locations on the lower limb.

difference between the IMU sensors placed on the the lateral thigh and back. The knee joint angle was represented by the sagittal plane angular difference between the IMU sensors on the lateral shank and lateral thigh. Similarly, the ankle joint angle corresponded to the sagittal plane angular difference between the IMU sensors located on the dorsum of the foot and the lateral shank.

2.1.5. Experimental protocol

In this experiment, a customized 30° ramp and a four-level staircase were used. As shown in Fig. 5, each participant was required to complete three gait modes: level walking, ramp ascent, and stair ascent. The

experimental protocol was as follows: participants walked on level ground four times, each session lasting 4 min with a 1-minute rest interval to prevent muscle fatigue. Ramp ascent was performed four times, each lasting 5 min with a 1-minute rest. Stair ascent was also completed four times, each lasting 5 min with a 1-minute rest period.

2.2. Signal processing

2.2.1. Signal filtering

sEMG signals represent the temporal and spatial superposition of motor unit action potentials (MUAPs) detected by surface electrodes.

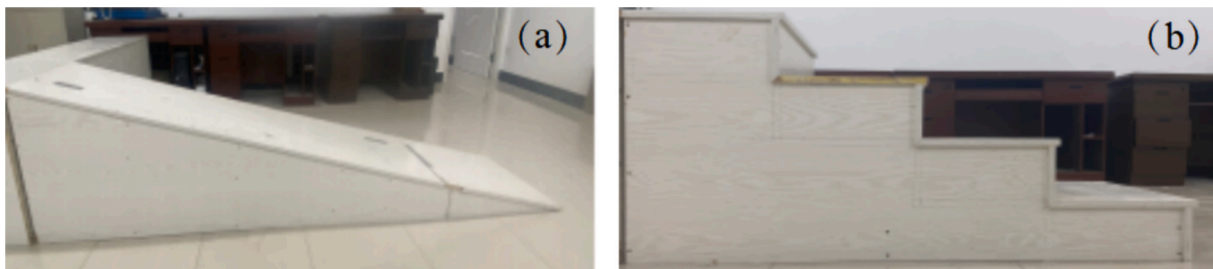


Fig. 5. Experimental walking platform. (a) ramp; (b) stairs.

Due to the short-term stationarity of sEMG signals, noise components are inevitably introduced during signal superposition, primarily consisting of powerline interference and motion artifacts. Failure to remove these noise components can lead to signal distortion, in severe cases, the effective signal may be completely obscured by noise, preventing extraction of valid features for input into recognition algorithms. To reduce the influence of these noise components, sEMG signals were band-pass filtered between 5 to 240 Hz. sEMG and IMU signals have different sampling rates, both were resampled at 500 Hz. Fig. 6 shows the preprocessed sEMG signal during level walking for one participant.

2.2.2. Data Windowing

Using a sliding window approach, the sEMG signal sequence was segmented into a series of overlapping or non-overlapping windows. Feature extraction and analysis were performed within each window to obtain time-varying signal characteristics. As shown in Fig. 7, a window length of 10 sampling points (20 ms) was used, with a 20% overlap (4 ms) to obtain the subsequent window. Through this sliding window method, features were extracted from seven lower limb muscles within each window.

Past data were used to predict future joint angles. Features extracted from sliding windows 1 to 10 were used to predict the joint angle in window 11 (as shown in Fig. 8), features from windows 2 to 11 were used to predict the angle in window 12, and so forth. sEMG signals are generated within the first 200 ms of movement formation. In this study, sequential features from 10 sliding windows with 20% overlap were used to predict subsequent joint angles, meeting the temporal requirements of the prediction task.

2.2.3. Feature extraction

To reduce computational complexity, time-domain features were used for analysis. Six features were extracted: mean absolute value (MAV), waveform length (WL), third quartile (Q3), root mean square (RMS), weighted mean absolute value (WMAV), and variance (VAR). These features effectively characterize the state of muscle activity.

(1) MAV.

MAV reflects the degree of deviation of the sEMG signal curve from the horizontal axis and can represent muscle activity. This feature is estimated as the average of the absolute values of the signal within a window containing N samples. The formula for MAV is given as follows:

$$MAV = \frac{1}{N} \sum_{i=1}^N |x_i| \quad (1)$$

Where, x_i denotes the amplitude of the sEMG signal at the i^{th} sampling point, and N represents the total number of samples within the analysis window.

(2) WL.

WL is a measure of the complexity of the sEMG signal, defined as the cumulative length of the signal waveform over a given time interval. The formula for WL is as follows:

$$WL = \sum_{i=1}^{N-1} |x_{i+1} - x_i| \quad (2)$$

(3) Q3.

Quartiles are statistical values that divide a dataset, ordered from minimum to maximum, into four equal parts by three cutoff points. In this study, Q3, also known as the upper quartile, was used as a feature. It represents the 75th percentile of the data sorted in ascending order. The formula is as follows:

$$Q_3 = \text{quantile}(x, 0.75) \quad (3)$$

Where, x denotes the dataset of signal values.

(4) RMS.

RMS is an effective method for analyzing sEMG signal amplitude and muscle activation patterns. It characterizes the dynamic behavior of neuromuscular activity during gait, while providing insights into the relative contributions of specific muscles. RMS is calculated as the square root of the mean of the squared values within a defined analysis window, representing the effective value of the sEMG signal during that period. The formula is given as follows:

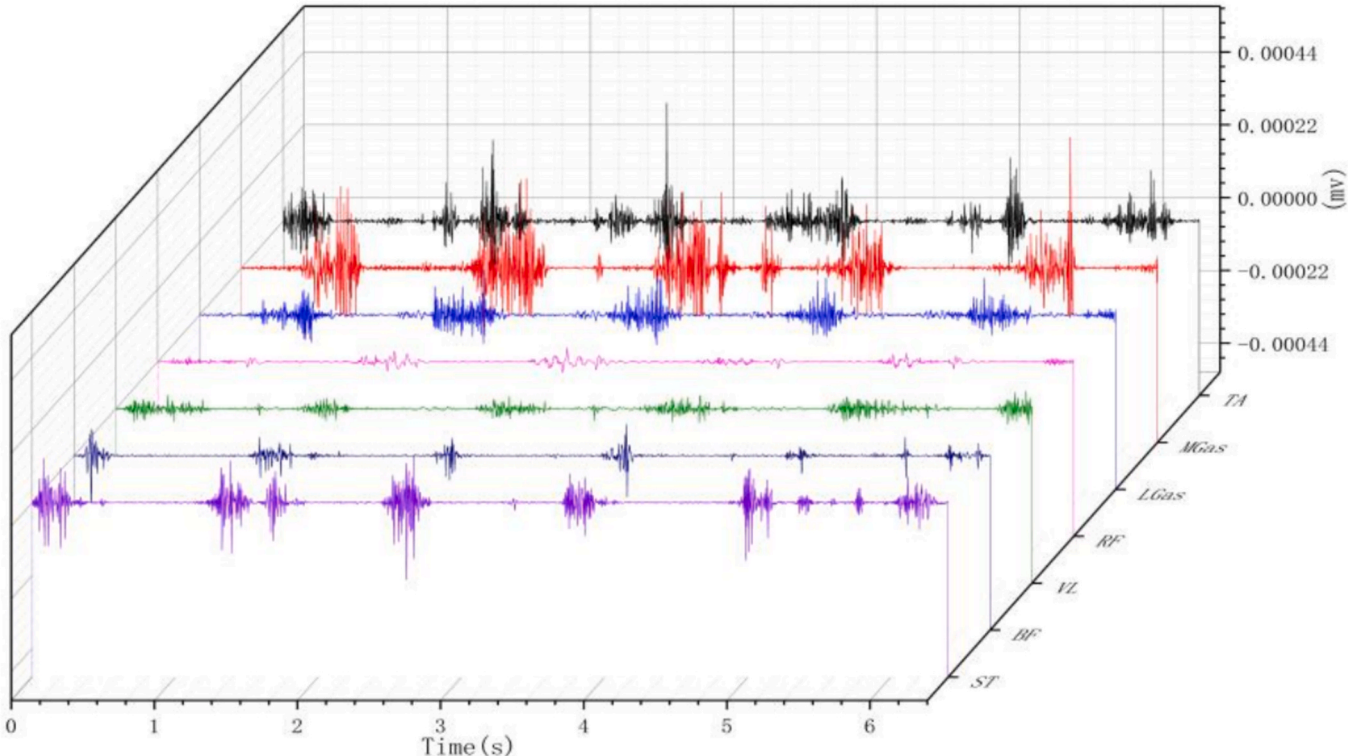


Fig. 6. Preprocessed sEMG signal during level walking.

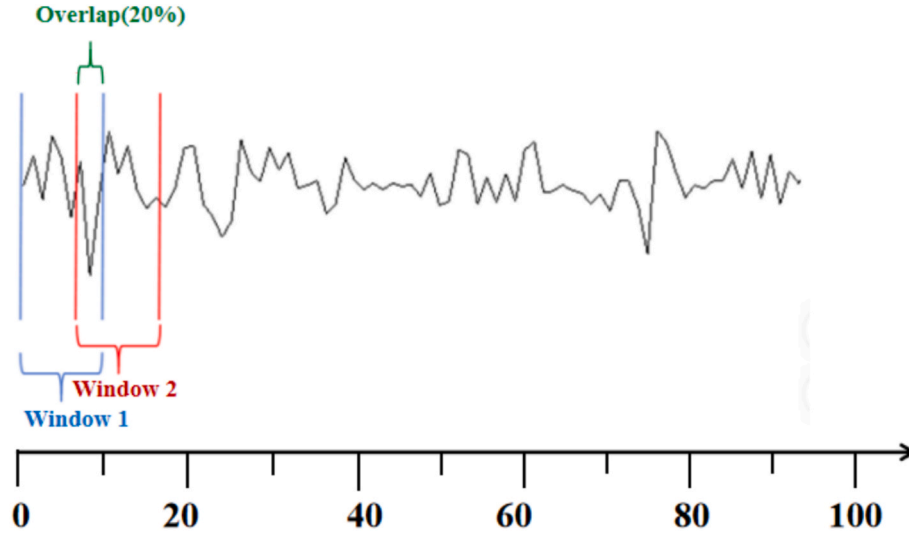


Fig. 7. Sliding window method.

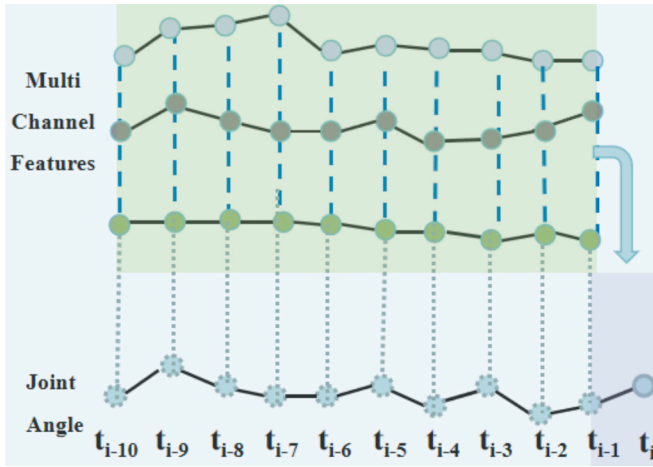


Fig. 8. Sliding window prediction method.

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (4)$$

(5) WMAV.

WMAV assigns different weights to different parts of the signal. If the index is located within the middle portion of the signal window (between 25% and 75% of the window length), a weight of 1 is assigned; otherwise, a weight of 0.5 is applied. The formula is given as follows:

$$WMAV = \frac{\sum_{i=1}^N |x_i| \cdot w_i}{\sum_{i=1}^N w_i} \quad (5)$$

$$w_i = \begin{cases} 1, & \text{if } [0.25 \times N] < i < [0.75 \times N] \\ 0.5, & \text{otherwise} \end{cases}$$

(6) VAR.

VAR measures the degree of deviation of a signal from its mean, providing information about the signal's fluctuation or variability. The formula is given as follows:

$$VAR = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (6)$$

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

Where, $\bar{x}$ denotes the mean value of the corresponding signal.

2.2.4. Joint angle calculation

As previously described, the true joint angles were obtained using acceleration and angular velocity signals from four IMU sensors, serving as the ground truth for joint angle prediction.

Attitude angles describe the orientation of an object in a three-dimensional coordinate system and are commonly represented using either Euler angles or quaternions. Quaternions are used to describe rotations in space and offer several advantages over Euler angles: they require less computational effort, avoid the gimbal lock problem, and enable more efficient and accurate attitude estimation. By using quaternions, the efficiency of coordinate transformation and attitude estimation can be improved, and the results can ultimately be converted into Euler angles. The Madgwick algorithm based on quaternion fusion was used for attitude estimation. The basic steps include: initializing the quaternion, integrating gyroscope data to update the quaternion, computing the accelerometer error, calculating the gradient for quaternion correction, normalizing the corrected quaternion, and outputting the final quaternion [41]. The resulting quaternion is then used to obtain its components, which are converted into a rotation matrix and subsequently transformed into Euler angles [42].

Following the aforementioned method, the joint angle was calculated as the difference between the angles measured by two IMU sensors. It was calculated that the standard deviations of the knee joint angles during level walking (LW), ramp ascent (RA), and stair ascent (SA) are 16.8°, 14.0°, and 22.6°, respectively; the corresponding values for the ankle joint are 11.5°, 8.1°, and 8.2°. The joint angle variation curves of one participant under the three gait modes are presented in Fig. 9. The interval between two dashed lines represents one gait cycle, which is defined as the time interval between two heel strikes of the same foot. Each subplot presents the signal variations across two consecutive gait cycles.

In addition, to align with the sEMG sliding window of 10 data samples, the joint angles were averaged using a sliding window of the same length (10 samples, corresponding to 20 ms), with a step size of 8 samples (20% overlap).

2.3. Joint angle Prediction

2.3.1 feature matrix

During the experimental phase, sEMG signals were collected from a total of seven muscles in the thigh and shank regions. For feature extraction, six features were computed for each sEMG channel. The features extracted from the previous 10 windows were used to predict

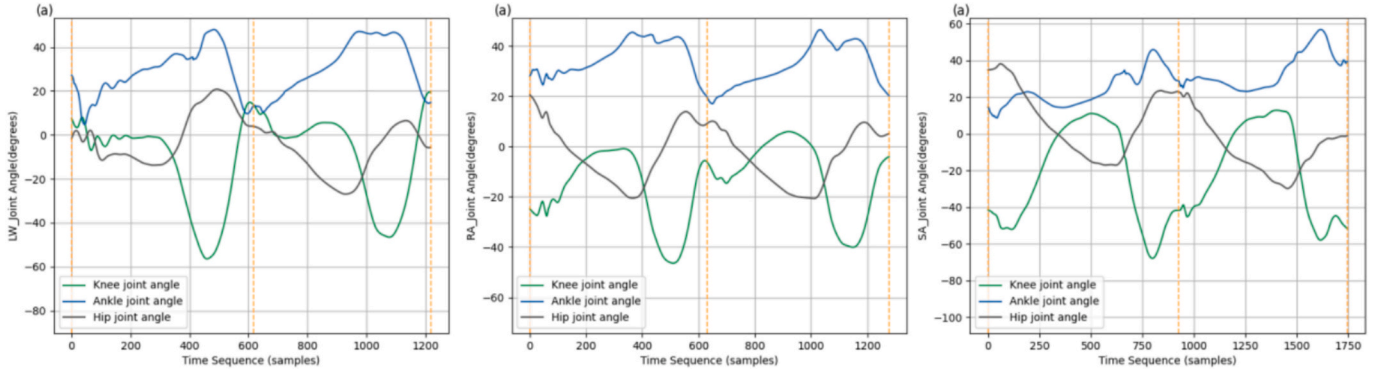


Fig. 9. Joint angle in three gait modes. (a) Level walking; (b) Ramp ascent; (c) Stair ascent.

the features in the 11th window. To investigate the predictive performance of unimodal multi-channel sEMG feature, unimodal few-channel sEMG feature, and sEMG-joint angle features, three feature combination matrices were selected as inputs for the regression prediction models.

- (1) Feature Matrix I: Feature set comprising sEMG signal features from seven muscles in the thigh and shank, with the total number of features as follows: $7 \times 6 \times 10$.
- (2) Feature Matrix II: Feature set comprising sEMG signal features from four thigh muscles, with the total number of features as follows: $4 \times 6 \times 10$.
- (3) Feature Matrix III: Feature set comprising sEMG signal features from four thigh muscles combined with hip joint angle, with the total number of features as follows: $4 \times 6 \times 10 + 10$.

Feature Matrix II primarily considers that individuals with lower-leg amputations constitute a large proportion of the mobility-impaired population. With their shank muscles absent, four thigh muscles were selected. Feature Matrix III uses a fusion approach, combining features from four thigh sEMG channels with hip joint angle data, aiming to improve prediction performance.

2.3.1. Normalization

A feature matrix includes various sEMG features and may also incorporate hip joint angle. Due to differences in units and scales among these features, this disparity can increase the difficulty of model training and adversely affect the model's generalization ability. Normalizing the input feature matrix ensures that all data share the same scale, which improves the training speed, generalization capability, and stability of the model. Normalization transforms the data distribution to a standard normal distribution with a mean of 0 and a standard deviation of 1. This method does not alter the shape of the data distribution but only adjusts the scale. The normalization formula is shown in Eq. (7).

$$x' = \frac{x - \mu}{\sigma} \quad (7)$$

Where, x represents the raw data point, μ is the mean of the data, and σ is the standard deviation of the data.

2.3.2. LSTM based prediction model

The LSTM architecture was adopted for joint angle prediction due to its proven capability in modeling temporal dependencies in biomechanical signals. LSTM is an improved version of the RNN, retaining the classic chain structure of RNNs and possessing a certain memory capability. To overcome the long-term dependency problem inherent in traditional RNNs, LSTM introduces a unique memory cell mechanism. This mechanism selectively stores previous state information and dynamically controls the retention and forgetting of information through a sophisticated gating system, including the input gate, forget

gate, and output gate [43]. As illustrated in Fig. 10.

(1) Forget gate.

The forget gate determines which information to discard from the cell state. It outputs values between 0 and 1 through a sigmoid activation function, representing the degree of information retention. The forget gate output is given by Eq. (8).

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (8)$$

Where, f_t denotes the forget gate output, σ represents the sigmoid activation function, W_f is the weight matrix of the forget gate, h_{t-1} is the hidden state from the previous time step, x_t is the current input, and b_f is the bias term of the forget gate.

(2) Input gate.

The input gate determines how much of the current input information is written into the cell state. The output of the input gate, i_t is calculated as shown in Eq. (9), and the candidate cell state values, $\tilde{C}_t$ is generated according to Eq. (10).

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (9)$$

$$\tilde{C}_t = \tanh(W_c[h_t, x_t] + b_c) \quad (10)$$

Where, W_i and W_c are the weight matrices for the input gate and candidate values, respectively. b_i and b_c are the bias terms for the input gate and candidate values, respectively, and $\tanh$ denotes the hyperbolic tangent activation function. The updated cell state C_t , which incorporates the filtered new information from the current input, is computed as shown in Eq. (11).

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (11)$$

(3) Output gate.

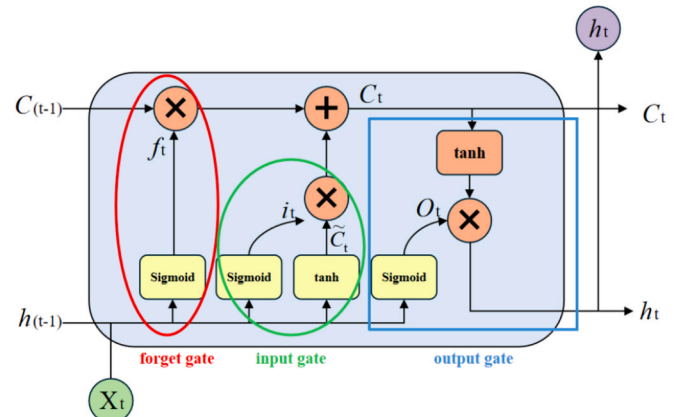


Fig. 10. The gated architecture of LSTM.

The output gate determines the value of the next hidden state, which contains information from the previous time step and can be used to predict the next moment. The output gate functions by weighting the information stored in the memory cell together with the current input to generate the output. The output of the output gate, O_t , is calculated as shown in Eqs. (12) and (13).

$$O_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (12)$$

$$h_t = O_t \cdot \tanh(C_t) \quad (13)$$

Where, W_o is the weight matrix of the output gate, b_o is the bias term of the output gate, h_t represents the hidden state at the current time step.

2.3.3. NRBO-LSTM model optimization

In most existing studies, the selection of model parameters relies on manual experience, which is inevitably influenced by subjective factors. To address this limitation, we employ NRBO to optimize the LSTM model parameters for achieving global optimality. NRBO represents a novel population-based metaheuristic algorithm designed for continuous optimization problems. Its core philosophy integrates the Newton-Raphson Search Rule (NRSR) with the Trap Avoidance Operator (TAO) to enhance optimization efficiency while preventing convergence to local optima [44]. Specifically, the NRSR accelerates convergence and improves search capability through gradient-informed updates. The TAO prevents premature convergence by introducing strategic stochastic perturbations.

In NRBO, the NRSR is implemented through the following steps: first, the first- and second-order derivatives of the objective function are approximated using a Taylor series expansion. As the objective function in optimization problems may be non-differentiable, NRSR approximates the derivatives using finite difference methods, as shown in Eqs. (14) and (15).

$$f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{2\Delta x} \quad (14)$$

$$f''(x) \approx \frac{f(x + \Delta x) + f(x - \Delta x) - 2f(x)}{(\Delta x)^2} \quad (15)$$

Subsequently, the position of the current solution is updated using the derivative expressions mentioned above, as shown in Eq. (16).

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)} \quad (16)$$

To enhance the algorithm's stochasticity and global search capability, the NRSR incorporates a stochastic factor as shown in Eq. (17).

$$x_{n+1} = x_n - \text{randn} \times \frac{f'(x_n)}{f''(x_n)} \quad (17)$$

Where, randn is a random number drawn from a normal distribution with a mean of 0 and a standard deviation of 1.

TAO introduces random perturbations to enable the algorithm to escape local optima and continue exploring potentially better solutions, as shown in Eq. (18):

$$x_{\text{new}} = x_{\text{current}} + \theta \times (\mu_1 \times x_{\text{best}} - \mu_2 \times x_{\text{current}}) \quad (18)$$

Where, θ is the random perturbation factor, μ_1 and μ_2 are random coefficients, x_{best} represents the current best solution. If the new candidate solution outperforms the current one, it is accepted; otherwise, the current solution is retained.

The NRBO algorithm is employed to determine the optimal hyperparameters for the LSTM model, including: the number of hidden units, initial learning rate, and L2 regularization coefficient. The implementation comprises the following steps:

Initialization Phase: Randomly generate an initial population of candidate solutions; Configure algorithm parameters (Population size, Maximum iteration count, Lower and upper bounds for each parameter).

Iterative Optimization Phase: For each iteration, update all hyperparameter combinations using the NRSR; Apply the TAO to introduce strategic perturbations, preventing local optima entrapment; Evaluate the fitness of each hyperparameter set; Update the global optimal solution.

Termination Condition: Repeat the optimization cycle until either the maximum iteration count is reached, or the convergence criteria are satisfied.

2.3.4. Performance evaluation

To ensure robust generalization performance of the model, the feature matrix was partitioned into training, and test sets [45]. 90% of the data were allocated to the training set, and the remaining 10% to the test set. Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2) were adopted as evaluation metrics for joint angle prediction. These three metrics are widely used and recognized in the literature for assessing the accuracy of human joint angle estimations [46–49].

(1) RMSE

Root Mean Square Error (RMSE) is one of the most commonly used metrics for measuring the discrepancy between predicted and true values. It is calculated as the square root of the average of the squared differences between predictions and ground truths. A smaller RMSE indicates higher predictive accuracy of the model. The formula is shown in Eq. (19).

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (19)$$

Where, y_i denotes the ground truth of the i^{th} sample, $\hat{y}_i$ represents the predicted value, and n is the total number of samples.

(2) MAE

Mean Absolute Error (MAE) is used to measure the difference between the model's predictions and the true values. A smaller MAE indicates higher predictive accuracy. The calculation formula is presented in Eq. (20).

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (20)$$

(3) R^2

The coefficient of determination, denoted as R^2 , is commonly used to evaluate the predictive performance of regression models. Its value ranges between 0 and 1, where values closer to 1 indicate higher predictive accuracy. The calculation formulas are presented in Eqs. 21, 22, and 23.

$$R^2 = 1 - \frac{SSE}{SST} \quad (21)$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (22)$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 \quad (23)$$

In the formula, SSE denotes the residual sum of squares between the observed and predicted values, reflecting the discrepancy between the model's predictions and the true values. A smaller SSE indicates better goodness of fit of the model to the data. SST represents the total sum of squares between the observed values and their mean, reflecting the variability of the true values. A larger SST corresponds to greater variability in the data, providing a stronger basis for the model to explain the variance and achieve a better fit.

3. Results

3.1. Prediction results based on LSTM

3.1.1. Model structure and parameters

Fig. 11 and Table 1 respectively illustrate the LSTM network architecture and its parameter configuration. Parameters such as batch size and epochs were manually tuned and experimentally validated to achieve optimal model performance.

3.1.2. Unimodal multi- sEMG features

Table 2 summarizes the RMSE, MAE, and R^2 for knee and ankle joint angles across three gait patterns by using 7ch multi-channel sEMG features. To minimize random variability, all reported results represent averages of 10 repeated runs per participant, consistent with our data processing protocol throughout this study.

3.1.3. Unimodal few-sEMG features

Prediction of knee and ankle joint angles was performed using sEMG features from four thigh muscles, with average results across all participants under different gait modes presented in Table 3. Comparative analysis in Table 2 demonstrates degraded performance across all metrics when compared to 7-channel EMG features, with increased error values (RMSE, MAE) and reduced R^2 .

3.1.4. sEMG-joint angle features

To improve joint angle predictions using only thigh muscle signals, we introduced a method that combines hip joint angles with four-channel thigh sEMG features. The rationale and effectiveness of this approach have been validated in previous studies [2]. Table 4 summarizes the results for knee and ankle joint angles across three gait patterns.

To compare the predictive performance across different feature matrices, Fig. 12 displays histograms of the RMSE and MAE results for both knee and ankle joints. To further evaluate the goodness-of-fit of the regression models, the R^2 values were categorized into four intervals: [0,0.3), [0.3,0.6), [0.6,0.8), and [0.8,1.0], corresponding to poor, general, good, and excellent prediction performance, respectively. Fig. 13 employs stacked histograms to visualize and analyze the proportional distribution of R^2 values across these four intervals.

The above results demonstrate that compared to using either 7-channel or 4-channel sEMG features alone, the fused feature approach significantly improves joint angle prediction accuracy. However, the current prediction performance still has room for optimization. To further enhance prediction accuracy, this study introduces the NRBO algorithm into the LSTM network model to improve its generalization capability and stability.

3.1.5. Statistical analysis of feature matrices

To further validate these results, we conducted statistical analyses. Given the independence of the data, we performed significance tests to compare the predictive performance (RMSE, MAE, and R^2) of different

Table 1

Hyperparameters of the Neural Networks.

Name	Value
Batch_size	32
Epochs	300
Optimizer	Adam
Number of Nodes	30
Input Size	10

feature matrices under the LSTM model. Prior to this, normality and homogeneity of variance were first assessed: If the data met both normality and homogeneity of variance, a one-way ANOVA was used. If the data met normality but not homogeneity of variance, Welch's ANOVA was employed. If the data violated normality, the Kruskal-Wallis test was applied instead. Only if the omnibus test was significant ($p < 0.05$), post hoc tests were performed to determine pairwise significance levels. The analysis results are presented in Table 5.

3.2. Model performance Comparison

3.2.1. Prediction results

To systematically evaluate the performance of the NRBO algorithm, we employs a Multi-Head Attention LSTM (MHA-LSTM) as a comparative model for training and prediction. The basic LSTM model adopts the architecture illustrated in Fig. 11, while the MHA-LSTM model utilizes the enhanced structure depicted in Fig. 14. Compared to the basic LSTM, the MHA-LSTM introduces a multi-head attention mechanism, increasing the complexity of the network hierarchy and significantly expanding the input node scale of the LSTM. By comparatively analyzing the predictive performance of these two LSTM architectures, this study explores the influence mechanisms of different network structures on prediction outcomes. Furthermore, the NRBO algorithm is integrated with the basic LSTM model to investigate the effectiveness of improving prediction performance through parameter optimization while maintaining the original hierarchical structure of the model.

The prediction results of knee and ankle joint angles based on sEMG features from 4 channels of the thigh combined with the hip joint angle, are presented in Tables 6 and 7, respectively. Histograms of the RMSE and MAE results, along with the proportional distribution of R^2 values, are displayed in Figs. 15 and 16. Table 8.

Examining subject 1 as a representative case, the knee and ankle joint angle prediction results of three LSTM models using fused sEMG features and hip joint angles are presented in Figs. 17 and 18.

3.2.2. Statistical analysis of models

The statistical analysis of the prediction performance of different models using the fused feature matrix is presented in Tables 8.

4. Discussion

In terms of joint angle prediction, numerous studies have focused

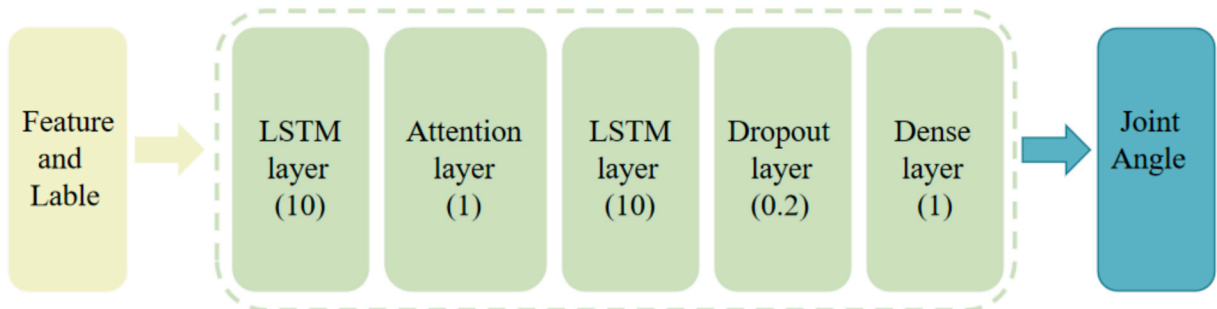


Fig. 11. Structure of LSTM neural network.

Table 2

Prediction performance for all participants (7ch sEMG).

Subject	Metric	LW_Knee	LW_Ankle	RA_Knee	RA_Ankle	SA_Knee	SA_Ankle
01	RMSE	6.6585	4.6049	5.9027	4.5191	6.8531	7.1142
	MAE	4.9973	3.4640	4.3880	3.3299	5.4085	5.2894
	R ²	0.8907	0.8854	0.8211	0.7280	0.9197	0.5924
02	RMSE	4.9275	3.6389	6.3143	3.5914	6.999	3.9216
	MAE	3.5802	2.8527	5.0569	2.6539	5.2838	3.0720
	R ²	0.7866	0.8022	0.7697	0.6317	0.9165	0.4565
03	RMSE	6.8381	4.3014	7.0633	3.8474	10.6033	6.0878
	MAE	5.0842	3.2056	5.2964	2.8787	8.4569	4.3971
	R ²	0.8645	0.8763	0.7526	0.7291	0.7838	0.5750
04	RMSE	5.4742	3.7378	5.4141	3.3753	6.2751	3.6306
	MAE	4.2261	2.8704	4.1289	2.5859	4.7861	2.7022
	R ²	0.8873	0.8922	0.8870	0.8412	0.9243	0.7857
05	RMSE	5.8244	2.9333	6.3747	2.0993	9.1470	3.0783
	MAE	4.4187	2.2127	4.8200	1.6589	7.0969	2.3431
	R ²	0.8711	0.8708	0.7849	0.6358	0.8640	0.5174
06	RMSE	5.3258	3.1690	5.3663	3.7519	7.9826	3.8434
	MAE	4.0407	2.4774	3.9998	2.9092	5.9502	3.0279
	R ²	0.8919	0.8360	0.8193	0.5719	0.8300	0.4364
07	RMSE	6.7009	4.6699	5.3539	3.9355	8.0970	4.3324
	MAE	4.9490	3.6368	4.1667	3.0010	5.9636	3.1400
	R ²	0.8331	0.9111	0.7901	0.8677	0.8535	0.8026
08	RMSE	5.5408	3.6159	7.4107	5.4019	7.2587	4.5581
	MAE	4.2377	2.7389	5.4582	4.1396	5.4536	3.6690
	R ²	0.8837	0.8935	0.7104	0.7089	0.8610	0.6812
09	RMSE	5.9458	5.3691	6.7665	4.2520	6.9930	6.9008
	MAE	4.5648	3.9951	4.9008	2.9977	5.4090	5.7314
	R ²	0.9062	0.8427	0.8450	0.8104	0.9258	0.7269
10	RMSE	6.5723	5.4559	7.4223	4.2898	5.8882	4.5651
	MAE	4.8947	3.9790	5.7819	3.2376	4.5812	3.6000
	R ²	0.8330	0.6635	0.7004	0.7235	0.9192	0.5271
Avg $\pm$ Std	RMSE	5.98 $\pm$ 0.67	4.15 $\pm$ 0.87	6.34 $\pm$ 0.82	3.91 $\pm$ 0.85	7.58 $\pm$ 1.43	4.80 $\pm$ 1.41
	MAE	4.50 $\pm$ 0.49	3.14 $\pm$ 0.62	4.80 $\pm$ 0.61	2.94 $\pm$ 0.63	5.84 $\pm$ 1.15	3.70 $\pm$ 1.11
	R ²	0.86 $\pm$ 0.04	0.85 $\pm$ 0.07	0.79 $\pm$ 0.06	0.72 $\pm$ 0.10	0.88 $\pm$ 0.05	0.61 $\pm$ 0.13

Table 3

Average prediction results across all participants (4ch sEMG).

Gait	Metric	Knee	Ankle
LW	RMSE	9.7212 $\pm$ 1.5232	6.7417 $\pm$ 1.3283
	MAE	6.8605 $\pm$ 1.0625	4.9542 $\pm$ 0.9389
	R ²	0.6272 $\pm$ 0.1651	0.5970 $\pm$ 0.1729
RA	RMSE	7.4834 $\pm$ 1.0419	5.2409 $\pm$ 1.4604
	MAE	5.6773 $\pm$ 0.8088	3.8387 $\pm$ 1.0309
	R ²	0.6993 $\pm$ 0.1038	0.5381 $\pm$ 0.1174
SA	RMSE	10.9919 $\pm$ 2.8441	5.6849 $\pm$ 1.9492
	MAE	8.2519 $\pm$ 2.2302	4.3464 $\pm$ 1.5278
	R ²	0.7273 $\pm$ 0.1713	0.4831 $\pm$ 0.1488

exclusively on single-joint (knee) angle prediction [26,32,50–55], while some have investigated multiple lower-limb joints [25,36,38,39,56–58]. The present study specifically examines both ankle and knee joint predictions. Regarding signal sources, sEMG remains the predominant modality in current research. Existing studies most frequently utilize sEMG signals from four lower-limb muscles: RF, VL, BF, and GM. The present study collects sEMG data from seven lower-limb muscles, comprehensively covering these four key muscles while incorporating additional muscular inputs. We investigated joint angle prediction under three feature configurations: multi-channel EMG features, fewer-channel EMG features, and fused EMG-hip angle features, aiming to examine the impact of channel quantity and modality on prediction performance. Deep learning has been widely applied to joint angle prediction in this field [36,37,59]. A systematic review article investigating the literature on prediction of gait kinematics and kinetics found that LSTM networks demonstrated superior prediction accuracy [60]. Our study compares the predictive performance of three LSTM-based model variants. Most existing studies focus on joint angle prediction during level-ground walking or treadmill locomotion [25,26,38,39,50,53–55,58,61]. Specifically, Zhu et al. investigated knee

joint angle prediction during walking on a 30° ramp [23], while Xie et al. explored the prediction of knee and ankle joint angles during stair ascent and descent [62]. In contrast, the present study conducted the prediction of knee and ankle joint angles was performed across three gait scenarios: level walking, ramp ascent and stair ascent, using sEMG signals from seven muscles on the thigh and shank.

4.1. Performance analysis of different features

Results presented in Tables 2 demonstrate that the LSTM model achieves good performance in joint angle prediction. Considering individuals with below-knee amputations constitute a significant proportion of mobility-impaired populations and often have incomplete shank muscles, a reduced set of 4-channel thigh muscles was selected. As shown in Tables 3, compared with the 7-channel configuration, reducing the number of muscle channels from 7 to 4 leads to a decline in prediction performance for both knee and ankle joint angles across all three gait modes. This degradation is attributed to the partial loss of key information caused by fewer channels, which lowers the model's predictive capability.

Although the 7-channel configuration achieves higher accuracy, excessive sensors would: (1) impair human-machine interaction experience; (2) reduce user comfort; (3) fail to accommodate application scenarios for amputees. However, relying exclusively on 4-channel thigh sEMG signals also produces decrease of prediction accuracy. Then, we integrated 4-channel thigh sEMG features with hip joint angle data. The performance improvements achieved through this hybrid approach are quantitatively demonstrated in Table 4. As illustrated in Fig. 12 through comparative bar charts of RMSE and MAE metrics across three methodologies, the hybrid approach demonstrates superior performance characteristics, exhibiting both lower mean values for RMSE and MAE compared to other two methods and significantly tighter standard deviations, indicating enhanced prediction accuracy and improved

Table 4

Prediction performance for all participants (4ch sEMG + hip).

Subject	Metric	LW_Knee	LW_Ankle	RA_Knee	RA_Ankle	SA_Knee	SA_Ankle
01	RMSE	5.5133	4.3554	5.3550	4.0513	8.0609	5.6352
	MAE	4.2932	3.3898	3.9926	3.0330	6.2339	4.2909
	R ²	0.9251	0.8975	0.8527	0.7814	0.8889	0.7443
02	RMSE	4.2716	3.5858	4.9657	3.5112	6.2072	3.7984
	MAE	3.1421	2.8247	4.0943	2.7594	4.8943	2.9514
	R ²	0.8397	0.8079	0.8575	0.6480	0.9283	0.4904
03	RMSE	4.5344	4.0569	4.9845	4.1480	6.1776	4.9628
	MAE	3.4294	3.2272	3.6098	3.1741	4.5866	3.6522
	R ²	0.9404	0.8899	0.8768	0.6852	0.9265	0.7175
04	RMSE	4.8250	3.6631	4.7113	3.2737	4.8241	3.3620
	MAE	3.7605	2.8304	3.5019	2.5747	3.8336	2.4513
	R ²	0.9124	0.8965	0.9144	0.8506	0.9551	0.8163
05	RMSE	6.6568	3.4939	4.3396	1.9253	7.4478	2.9054
	MAE	4.6849	2.5892	3.3376	1.5332	6.0290	2.3327
	R ²	0.8316	0.8167	0.9003	0.6938	0.9100	0.5702
06	RMSE	5.3874	3.2698	4.9478	3.2901	7.2108	4.0425
	MAE	4.0642	2.5924	3.6092	2.6041	5.1599	3.1879
	R ²	0.8893	0.8254	0.8464	0.6708	0.8613	0.3766
07	RMSE	4.8675	4.8130	4.3833	3.6654	6.0565	4.5786
	MAE	3.7168	3.6297	3.4363	2.8004	4.6871	3.3239
	R ²	0.9119	0.9055	0.8594	0.8852	0.9180	0.7794
08	RMSE	5.1595	3.9343	5.8963	5.1530	6.7293	4.9572
	MAE	3.6644	2.8833	4.7266	4.0640	5.2151	4.1040
	R ²	0.8992	0.8739	0.8166	0.7352	0.8805	0.6229
09	RMSE	5.5266	5.0279	5.6306	4.5175	6.1864	6.3039
	MAE	4.2349	3.9372	4.2881	3.3327	4.6831	5.3851
	R ²	0.9190	0.8621	0.8928	0.7860	0.9417	0.7719
10	RMSE	6.6840	5.0091	7.1229	3.9690	6.9229	4.7194
	MAE	4.9504	3.7480	5.6420	2.9885	5.5779	3.6561
	R ²	0.8272	0.7165	0.7238	0.7633	0.8885	0.4949
Avg ± Std	RMSE	5.34 ± 0.81	4.12 ± 0.65	5.23 ± 0.83	3.75 ± 0.86	6.58 ± 0.90	4.53 ± 1.03
	MAE	3.99 ± 0.56	3.17 ± 0.49	4.02 ± 0.72	2.89 ± 0.64	5.09 ± 0.72	3.53 ± 0.91
	R ²	0.89 ± 0.04	0.85 ± 0.06	0.85 ± 0.05	0.75 ± 0.08	0.91 ± 0.03	0.64 ± 0.15

measurement consistency. The R² coefficient was categorized into four intervals, and Fig. 13 illustrates the proportional distribution of R² values across these ranges. The results demonstrate that the hybrid method exhibits significantly higher proportions in both the “good” and “excellent” intervals compared to the other two approaches, indicating substantial improvement in this parameter.

A number of studies have also adopted multi-source signals for joint angle prediction. Xie et al. utilized hip joint angle and 3-ch thigh sEMG signals as inputs, and implemented a Generalized Regression Neural Network (GRNN) to predict knee and ankle joint angles during walking [38]. Three metrics were employed for performance evaluation, namely RMSE, mean relative error (MRE), and correlation coefficient (γ). The corresponding values for the knee joint were 6.02°, 0.07 and 0.97, respectively, while those for the ankle joint reached 6.72°, 0.09 and 0.97. In contrast, the proposed method in our study, which takes 4-channel thigh sEMG features combined with hip joint angle data as inputs, yielded superior results under the walking scenario: the RMSE values were 5.34° for the knee joint and 4.12° for the ankle joint. Mobarak et al. adopted a similar approach, where hip joint angle and 3-ch thigh sEMG signals were also used as inputs, and a Least Squares-Support Vector Machine (LS-SVM) was applied to predict ankle joint angles during walking [61]. The results demonstrated that the incorporation of hip joint kinematic information can significantly enhance the accuracy of ankle joint angle estimation, which is consistent with the findings of the present study. Notably, although the sEMG electrode configuration in our study included one more channel than that in the above two studies, our work extended the experimental scenarios to ramp ascent and stair ascent in addition to level walking.

4.2. Performance analysis of different models

Building upon the fused feature approach, we further investigated joint angle prediction performance across three model architectures:

basic LSTM, MHA-LSTM and NRBO-LSTM, aiming to elucidate the impact of model selection on prediction outcomes. In the MHA-LSTM model, the incorporation of a multi-head attention mechanism not only elevates the architectural complexity of the network hierarchy but also substantially augments the input dimensionality of the LSTM layer. Predictably, the predictive performance is highly likely to surpass that of the basic LSTM model. However, the computational complexity and resource consumption of its training and prediction processes are relatively more significant. In the NRBO-LSTM, the NRBO algorithm, aiming to achieve rapid convergence while avoiding local optima, was used to optimize relevant parameters within the LSTM network. As shown in Tables 6,7 and Fig. 15, the MHA-LSTM generally outperforms the LSTM model, except for its inferior performance in ankle joint angle prediction during ramp ascent and stair ascent. In contrast, the NRBO-LSTM demonstrates more comprehensive improvements with lower RMSE and MAE. For knee joint angle prediction, after NRBO optimization, the average RMSE decreased by 22.2%, MAE by 22.5%, and R² increased by 4.7% across the three gait modes. For ankle joint angle prediction, the average RMSE and MAE decreased by 8.2% and 9.0%, respectively, while R² increased by 2.4%. In level walking, RMSE/MAE/R² of knee joint are $4.11 \pm 0.74^\circ / 3.04 \pm 0.6^\circ / 0.93 \pm 0.03$, RMSE/MAE/R² of ankle joint are $3.48 \pm 0.54^\circ / 2.66 \pm 0.41^\circ / 0.89 \pm 0.06$. In ramp ascent, RMSE/MAE/R² of knee joint are $4.33 \pm 0.76^\circ / 3.37 \pm 0.63^\circ / 0.9 \pm 0.04$, RMSE/MAE/R² of ankle joint are $3.6 \pm 0.78^\circ / 2.78 \pm 0.6^\circ / 0.76 \pm 0.08$. In stair ascent locomotion, RMSE/MAE/R² of knee joint are $4.91 \pm 1.0^\circ / 3.74 \pm 0.74^\circ / 0.95 \pm 0.02$, RMSE/MAE/R² of ankle joint are $4.3 \pm 0.7^\circ / 3.29 \pm 0.61^\circ / 0.64 \pm 0.21$. As illustrated in Fig. 16, the MHA-LSTM exhibits inferior R² results for ankle joint angle prediction compared to the LSTM model under both ramp ascent and level walking, the NRBO-LSTM demonstrates higher proportions in both the “good” and “excellent” intervals under all three gaits.

The predictive models adopted in existing studies cover categories such as neural networks, regression analysis, and decision tree

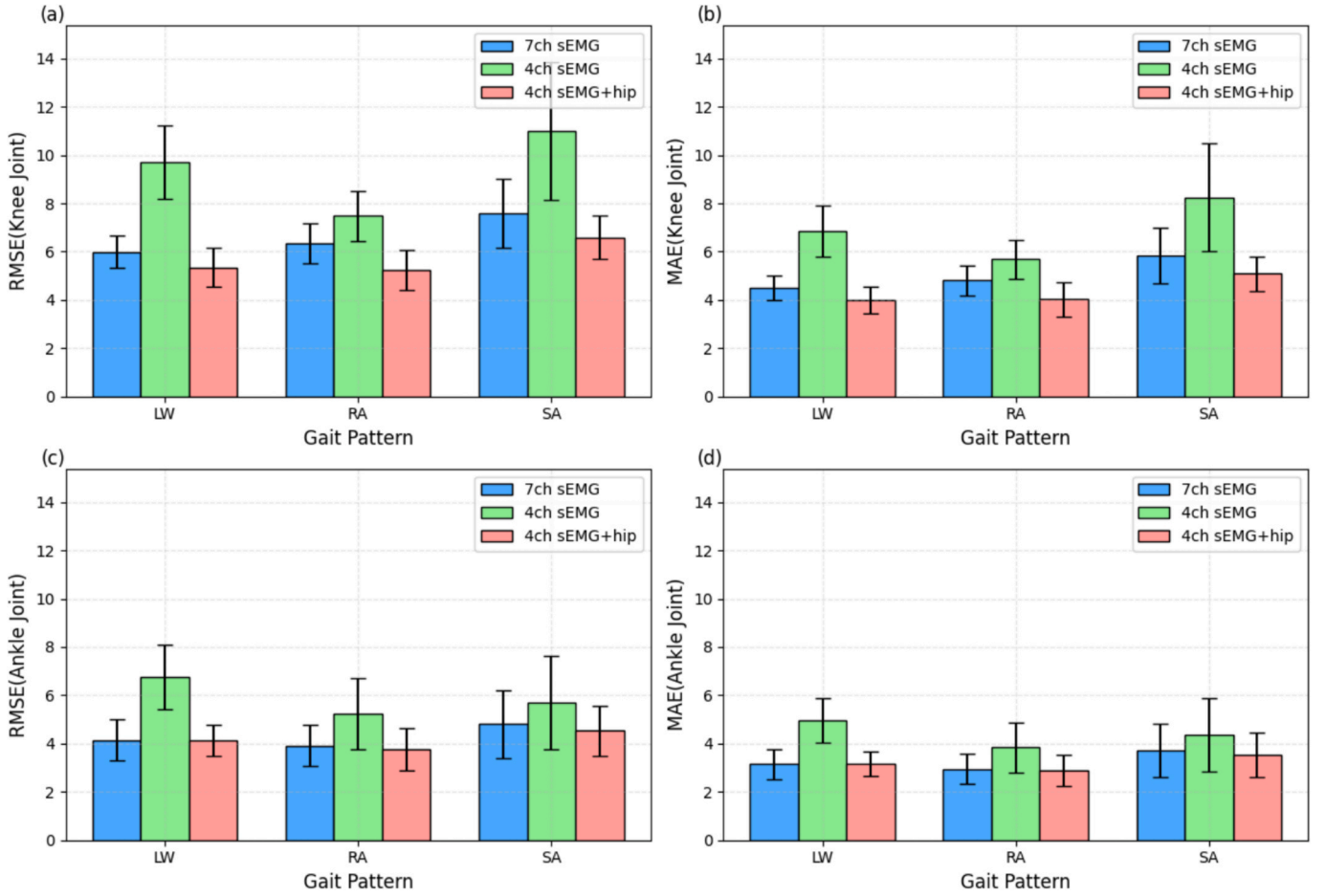


Fig. 12. RMSE and MAE results for three feature matrices. (a)RMSE result for the knee joint; (b)MAE result for the knee joint; (c)RMSE result for the ankle joint; (d) MAE result for the ankle joint.

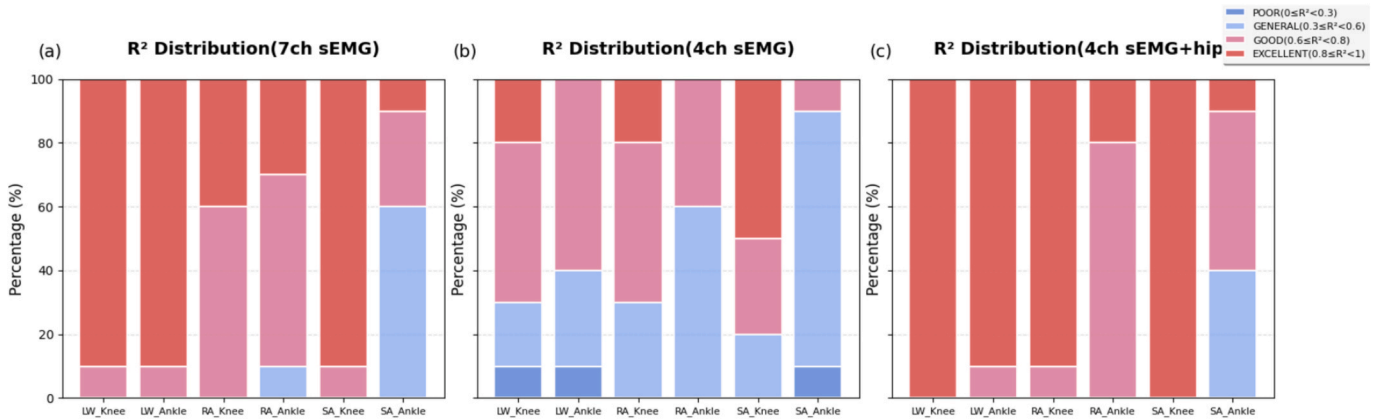


Fig. 13. R^2 distribution analysis. (a) 7ch sEMG features, (b) 4ch sEMG features, and (c) 4ch sEMG with hip features.

algorithms, among which the majority of literatures preferentially use neural networks and their variant models (e.g., CNN, DNN, RNN, LSTM) [23,26,38,39,50,54,62]. A review study further confirms that LSTM and CNN are superior neural network tools for constructing prediction systems based on sEMG signals [60]. This study selects the basic LSTM and its two variant models to compare the prediction performance differences under different model architectures. However, neither this study nor other related studies have formed a unified standard in terms of data processing methods and neural network model selection, resulting in the inability to determine the optimal predictive tool applicable to all

research scenarios at present.

4.3. Performance analysis of different joints and gaits

According to the results in Figs. 12 and 15, the RMSE and MAE of knee joint angles are consistently higher than those of ankle joints across all experimental conditions. Our results are consistent with the findings reported in Literature [39]. Based on the actual knee and ankle joint angles presented in Fig. 9, it can be inferred that a potential reason for this outcome is that the knee joint exhibits a more complex movement

Table 5

Significance analysis of prediction performance across different features.

Features	Metric	LW		RA		SA	
		Knee	Ankle	Knee	Ankle	Knee	Ankle
7ch VS 4ch	RMSE	***	***	*	*	*	N
	MAE	***	***	*	*	*	N
	R ²	**	**	N	***	N	N
7ch VS 4ch + hip	RMSE	N	N	*	N	N	N
	MAE	N	N	N	N	N	N
	R ²	N	N	*	N	N	N
4ch VS 4ch + hip	RMSE	***	***	***	*	**	N
	MAE	***	***	***	*	**	N
	R ²	**	**	**	***	*	N

trajectory with more significant fluctuations during gait, and such high movement complexity contributes to an increase in the model's prediction error. A similar finding was also reported in another study on multi-joint angle prediction, Lv et al. investigated the prediction of hip and knee joint angles during walking and found that the hip joint angle had a smaller fluctuation range than the knee joint angle, resulting in consistently lower RMSE and MAE values for the hip joint angle predictions compared with those for the knee joint [58]. The R² of ankle joint angle prediction is consistently lower than that of the knee joint, which may be attributed to the fact that ankle joint angles exhibit smaller standard deviations and thus their fluctuations during gait are less pronounced than those of the knee joint. In the calculation of R², on the assumption that the SSE is not particularly small, signals with smaller standard deviations tend to yield lower R² values.

Similar findings can be derived from different gait modes. Figs. 13 and 16 indicate that the R² values for ankle joint angles during ramp ascent and stair ascent are sub-optimal. During ramp ascent and stair ascent, the relatively limited movement between the shank and foot results in smaller ranges of ankle joint angle variation. In contrast, during level walking, ankle joint angles exhibit larger variations. This can explain why the prediction performance R² for ankle joint angles is better in level walking compared to ramp ascent and stair ascent. Most studies have conducted joint angle prediction during level walking or treadmill walking, a small number of studies have separately carried out experiments on stair ascent and descent [62] and ramp ascent [23], while no studies have yet analyzed the results of level walking, stair ascent, and ramp ascent simultaneously. The findings of this study show that joint angles exhibit distinct variation ranges across different gait modes. Signals with smaller variation ranges mean more stable data, which are easier for prediction models to model and thus yield lower RMSE and MAE, as minor variations can be predicted more accurately. Yet in some cases, good model fitting may not bring the expected high R² due to low inherent data variability. Therefore, RMSE and MAE reflect model prediction accuracy directly, whereas R² interpretation requires consideration of the data's contextual variability.

4.4. Analysis of statistical results

As evident from the figures 17 and 18, all prediction curves demonstrate consistent trends with the actual angular signal sequences, while the NRBO-LSTM model exhibits superior fitting performance in knee and ankle joint angle predictions.

The results of the significance analysis are presented in Table 5 and Table 8, respectively. In most experimental conditions, both 7ch features and fused features showed statistically significant improvements in prediction metrics compared to the 4ch features. However, no significant enhancement was observed for ankle joint metrics during stair ascent locomotion. The performance difference between 7ch features and fused features was generally non-significant across most test scenarios, with only selected metrics (RMSE and R²) showing significant improvements during ramp ascent locomotion. This outcome suggests that the fused feature approach can achieve prediction accuracy comparable to the full 7-channel configuration while utilizing fewer measurement channels, demonstrating its potential for more efficient

Table 6

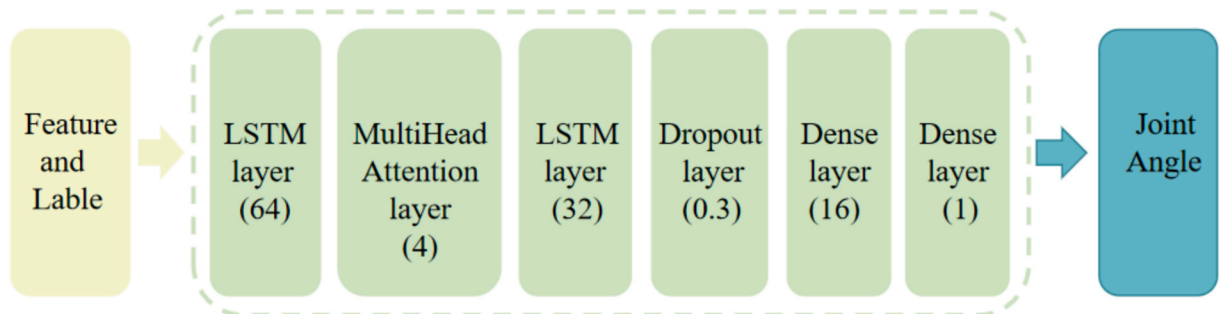
Prediction results of knee joint angles for three models.

Gait	Model	RMSE	MAE	R ²
LW	LSTM	5.3426 ± 0.8110	3.9941 ± 0.5606	0.8896 ± 0.0416
	MHA-LSTM	4.0391 ± 0.7240	3.0260 ± 0.5612	0.9324 ± 0.0389
	NRBO-LSTM	4.1121 ± 0.7442	3.0431 ± 0.5989	0.9317 ± 0.0345
RA	LSTM	5.2337 ± 0.8294	4.0238 ± 0.7165	0.8541 ± 0.0540
	MHA-LSTM	4.9243 ± 0.8721	3.7943 ± 0.6907	0.86689 ± 0.0519
	NRBO-LSTM	4.3279 ± 0.7559	3.3738 ± 0.6295	0.8996 ± 0.0371
SA	LSTM	6.5823 ± 0.9004	5.0901 ± 0.7186	0.9099 ± 0.0295
	MHA-LSTM	5.2753 ± 0.8406	3.8763 ± 0.5518	0.9406 ± 0.0256
	NRBO-LSTM	4.9132 ± 1.0025	3.7434 ± 0.7409	0.9482 ± 0.0245

Table 7

Prediction results of ankle joint angles for three models.

Gait	Model	RMSE	MAE	R ²
LW	LSTM	4.1209 ± 0.6496	3.1652 ± 0.4914	0.8492 ± 0.0588
	MHA-LSTM	3.7930 ± 0.5428	2.8595 ± 0.4272	0.8654 ± 0.0681
	NRBO-LSTM	3.4839 ± 0.5391	2.6646 ± 0.4086	0.8884 ± 0.0559
RA	LSTM	3.7504 ± 0.8620	2.8864 ± 0.6440	0.7500 ± 0.0783
	MHA-LSTM	4.0528 ± 0.8259	3.0743 ± 0.6174	0.6989 ± 0.1164
	NRBO-LSTM	3.6014 ± 0.7775	2.7836 ± 0.6035	0.7642 ± 0.0821
SA	LSTM	4.5265 ± 1.0302	3.5336 ± 0.9112	0.6385 ± 0.1501
	MHA-LSTM	4.8061 ± 0.7787	3.5618 ± 0.6791	0.5518 ± 0.2513
	NRBO-LSTM	4.3057 ± 0.6961	3.2888 ± 0.6116	0.6400 ± 0.2050

**Fig. 14.** Structure of MHA-LSTM neural network.

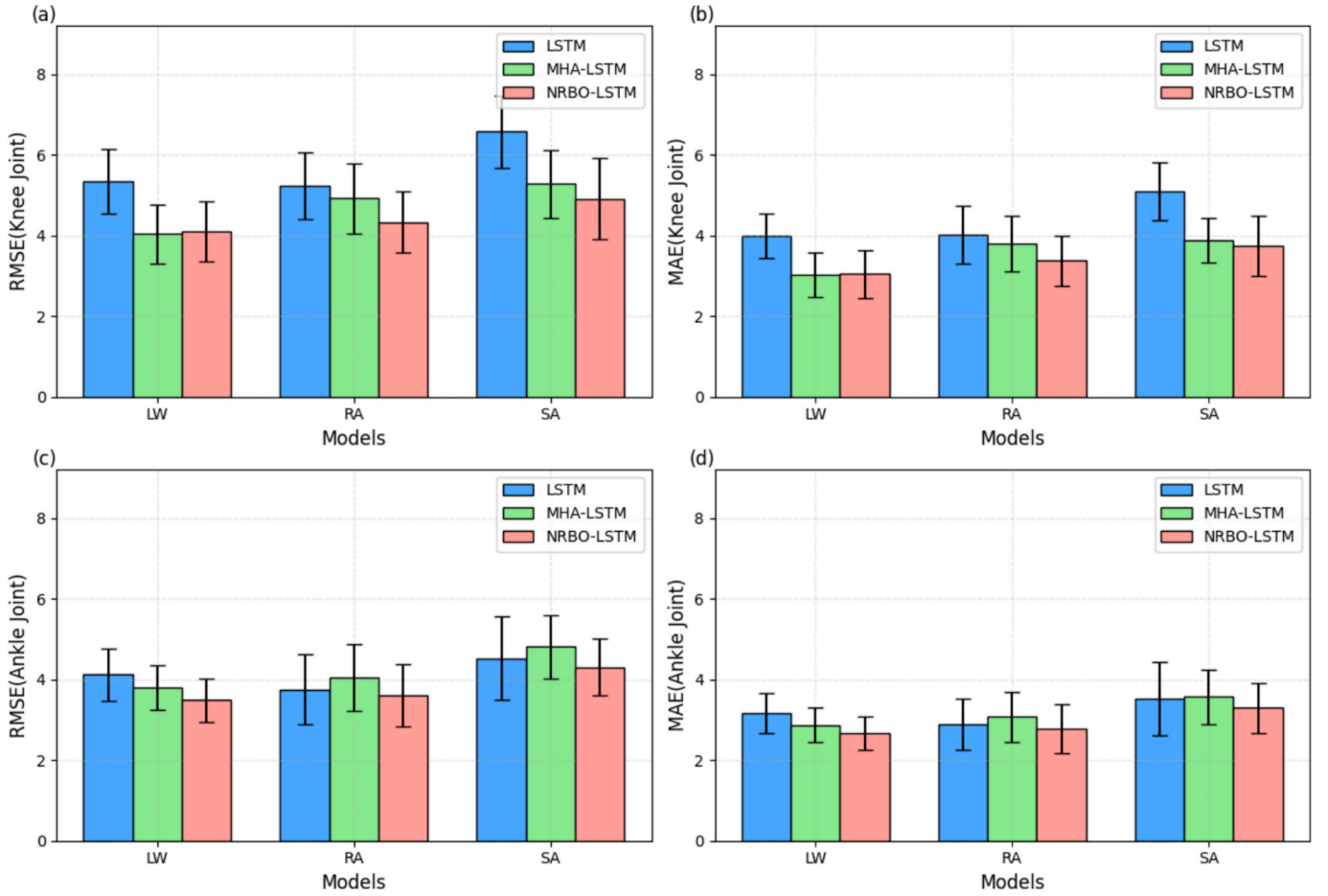


Fig. 15. RMSE and MAE results for three models. (a)RMSE result for the knee joint; (b)MAE result for the knee joint; (c)RMSE result for the ankle joint; (d)MAE result for the ankle joint.

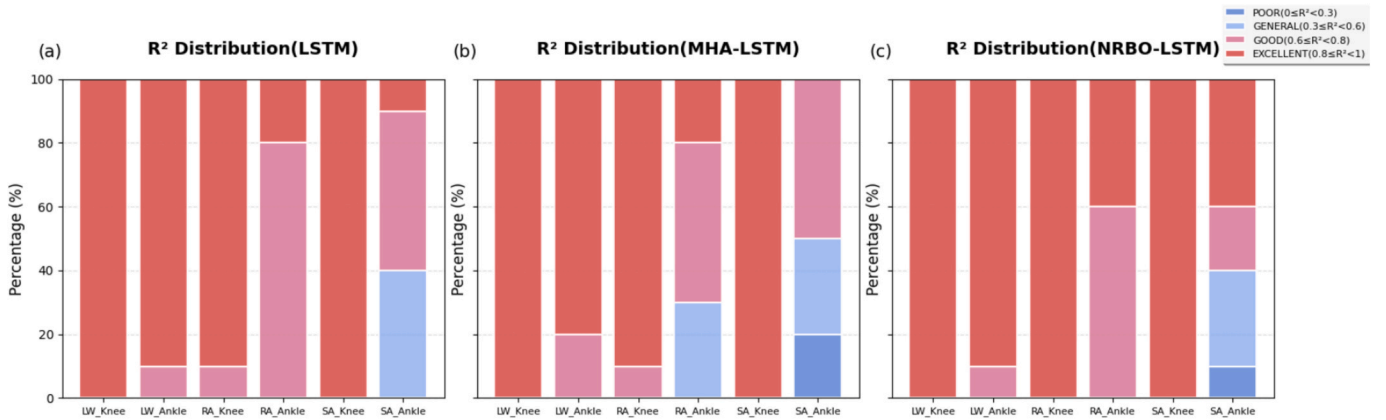


Fig. 16. R^2 distribution analysis. (a) LSTM; (b) MHA-LSTM; and (c) NRBO-LSTM.

motion analysis systems. In the significance analysis of model factors, LSTM and NRBO-LSTM exhibited some significant differences in performance, while LSTM and MHA-LSTM showed fewer significant differences. Besides, MHA-LSTM and NRBO-LSTM demonstrated no significant differences across all terrains and joint metrics. Based on the results shown in Figs. 15 and 16, it can be concluded that: For knee of level walking, ankle of level walking, and knee of stair ascent, the NRBO-LSTM model performs significantly better than the LSTM model but not significantly better than the MHA-LSTM model. For knee prediction of ramp ascent, the NRBO-LSTM model does not

significantly outperform the other two models. For ankle of ramp ascent and ankle of stair ascent predictions, the MHA model performs worse than the other two models, but the difference is not significant. These results indicate that the NRBO-LSTM approach enhances predictive accuracy through parameter optimization while maintaining the baseline LSTM's architectural complexity.

Additionally, the results reveal that increasing channel count (4ch vs 7ch) or incorporating hip data (4ch + hip) improves prediction performance. In particular, the prediction results of ankle joint angles obtained by the fusion method clearly confirm that even the incorporation of

Table 8

Significance analysis of prediction performance across different models.

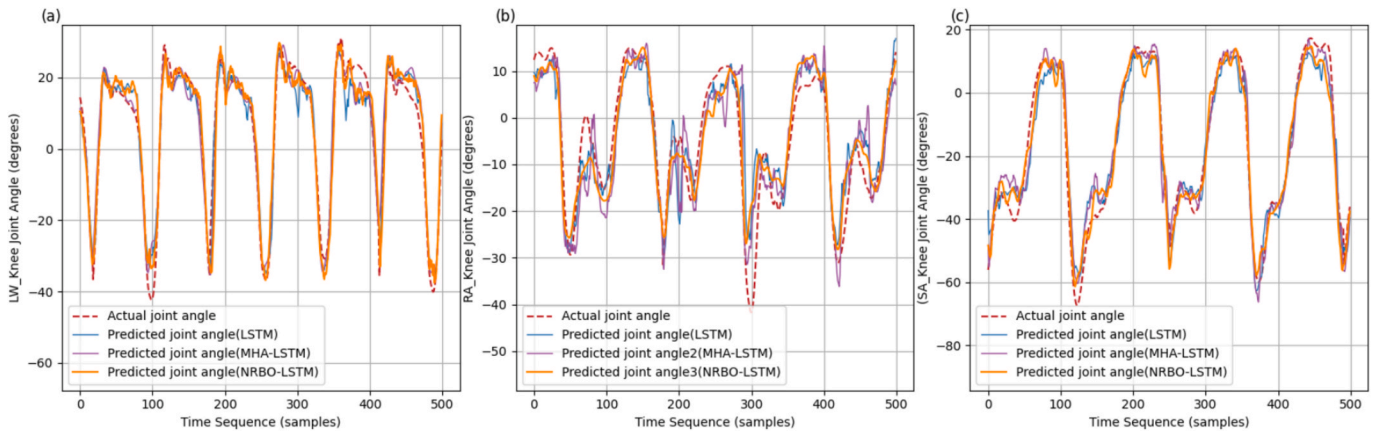
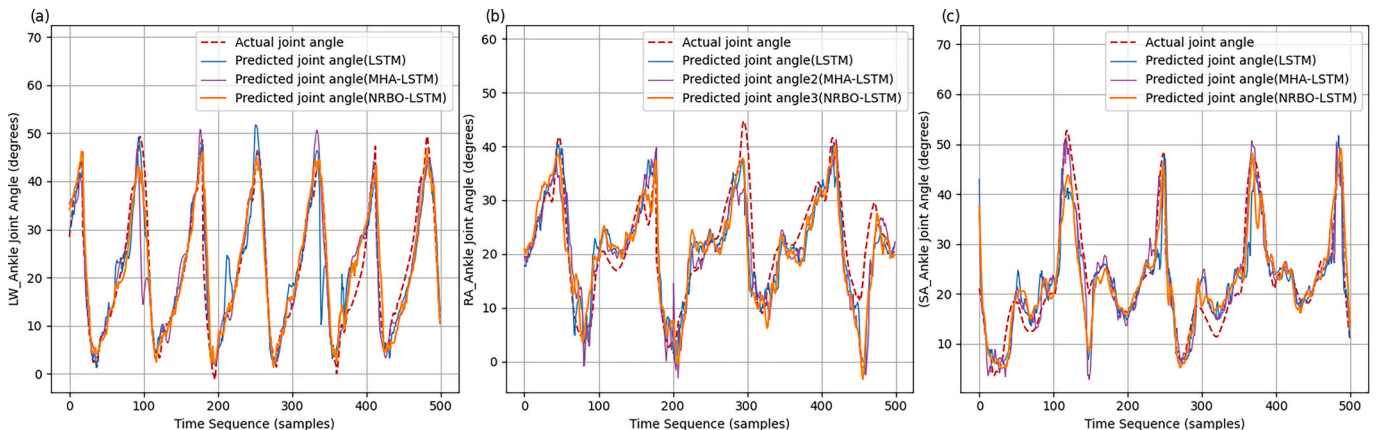
Model	Metric	LW		RA		SA	
		Knee	Ankle	Knee	Ankle	Knee	Ankle
LSTM VS MHA	RMSE	**	N	N	N	**	N
	MAE	**	N	N	N	**	N
	R ²	*	N	N	N	*	N
LSTM VS NRBO	RMSE	**	*	*	N	***	N
	MAE	**	*	N	N	***	N
	R ²	*	N	N	N	***	N
MHA VS NRBO	RMSE	N	N	N	N	N	N
	MAE	N	N	N	N	N	N
	R ²	N	N	N	N	N	N

kinematic information from the proximal hip joint, which has a weak functional relationship with the ankle joint, can effectively enhance the prediction performance of ankle joint angles. However, the enhancement effect on ankle joint angle prediction is relatively weaker under stair ascent. Regarding model comparisons, both MHA-LSTM and NRBO-LSTM demonstrate significantly better performance than LSTM for knee joint angle prediction in level walking and stair ascent, no significant improvement is observed in ramp ascent locomotion. For ankle joint angle prediction, performance differences across models are generally non-significant. These findings suggest that knee joint angle predictions benefit more from advanced methods and that terrain characteristics mediate the effectiveness of enhancement, highlighting the importance of context-specific optimization in prediction systems.

5. Conclusion

This paper proposes a method for predicting continuous knee and ankle joint angles. The study first collected 7-channel sEMG signals and 4-channel IMU data from 10 healthy subjects during level ground, ramp ascent, and stair ascent walking using Delsys wireless sEMG sensors. The sEMG signals underwent filtering, sliding window preprocessing and time-domain feature extraction, while the IMU signals were processed using quaternion-based joint angle calculation. Three input modalities were compared: single 7-channel sEMG signals, single 4-channel sEMG signals, and fused 4-channel sEMG with hip joint angle signals. Experimental results showed that the single 4-channel sEMG signals had the worst prediction performance, while no significant difference was observed between the single 7-channel sEMG signals and the fused 4-channel sEMG with hip joint angle signals, though the fused signal-based prediction model was more suitable for prosthetic control scenarios in amputees. The study also compared three prediction models: LSTM, MHA-LSTM and NRBO-LSTM. The results demonstrate that the NRBO-LSTM model not only achieves superior prediction performance but also exhibits greater lightweight characteristics, making it better suited for real-time control requirements.

This study provides a novel solution for prosthetic control in lower-limb amputees, but several limitations remain to be addressed. The dataset remains relatively small, and more participants should be recruited in future studies. The current dataset only includes signals from healthy subjects; future studies should incorporate data from amputees and individuals with neuromuscular disorders to improve the generalizability of the findings. To date, EEG-based joint angle prediction has been scarcely explored, and no studies have been conducted

**Fig. 17.** Knee Joint Angle Prediction Plots (4-ch sEMG + Hip). (a) Level walking; (b) Ramp ascent; (c) Stair ascent.**Fig. 18.** Ankle Joint Angle Prediction Plots (4-ch sEMG + Hip). (a) Level walking; (b) Ramp ascent; (c) Stair ascent.

on joint angle prediction using EMG-EEG fusion. Our future efforts will focus on advancing research in this direction.

Funding sources.

This research was partly funded by the National Natural Science Foundation of China (Grant Number: 81601586) and Fundamental Research Funds for the Central Universities, South-Central Minzu University (Grant Number: CZZ24015).

CRediT authorship contribution statement

Songbo Zhou: Writing – original draft, Methodology, Formal analysis. **Xinyi Ma:** Writing – original draft, Methodology, Formal analysis. **Yongxin Lv:** Formal analysis. **Na Ta:** Formal analysis. **Chaoming Luo:** Formal analysis. **Hui Zhou:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] L. Avellar, A. Leal-Junior, C. Marques, A. Frizzera, Performance Analysis of a Lower Limb Multi Joint Angle Sensor Using CYTOP Fiber: Influence of Light Source Wavelength and Angular Velocity Compensation, *Sensors* 20 (2) (2020), <https://doi.org/10.3390/s20020326>.
- [2] X. Liu, Continuous prediction of human knee joint angles and gait recognition during walking (Doctoral dissertation), China University of Petroleum, Beijing, 2023, <https://doi.org/10.27643/d.cnki.gsybu.2023.000016>.
- [3] D. Gafurov, E. Snekenes, Gait Recognition Using Wearable Motion Recording Sensors, *EURASIP Journal on Advances in Signal Processing* 2009 (1) (2009), <https://doi.org/10.1155/2009/415817>.
- [4] S. Sprager, M. Juric, Inertial sensor-based gait recognition: a review, *Sensors* 15 (9) (2015) 22089–22127, <https://doi.org/10.3390/s150922089>.
- [5] C. Morbidoni, A. Cucchiarelli, V. Agostini, M. Knaflitz, S. Fioretti, F. Di Nardo, Machine-learning-based prediction of gait events from EMG in cerebral palsy children, *IEEE transactions on Neural Systems and Rehabilitation Engineering* 29 (2021) 819–830, <https://doi.org/10.1109/TNSRE.2021.3076366>.
- [6] C. Morbidoni, A. Cucchiarelli, S. Fioretti, F. Di Nardo, A deep learning approach to EMG-based classification of gait phases during level ground walking, *Electronics* 8 (8) (2019) 894, <https://doi.org/10.3390/electronics8080894>.
- [7] P. Wei, R. Xie, R. Tang, C. Li, J. Kim, M. Wu, sEMG based gait phase recognition for children with spastic cerebral palsy, *Annals of Biomedical Engineering* 47 (1) (2018) 223–230, <https://doi.org/10.1007/s10439-018-02126-8>.
- [8] P. Terrier, Gait Recognition via Deep Learning of the Center-of-pressure Trajectory, *Applied Sciences* 10 (3) (2020) 774, <https://doi.org/10.3390/app10030774>.
- [9] J.B.D.M. Saunders, V.T. Inman, H.D. Eberhart, The major determinants in normal and pathological gait, *The Journal of Bone & Joint Surgery* 35 (3) (1953) 543–558, <https://doi.org/10.2106/00004623-195335030-00003>.
- [10] J. Taborri, E. Palermo, S. Rossi, P. Cappa, Gait Partitioning Methods: a Systematic Review, *Sensors* 16 (1) (2016) 66, <https://doi.org/10.3390/s16010066>.
- [11] A.D. Grant, Gait Analysis: Normal and Pathological Function, *JAMA* 304 (8) (2010) 907, <https://doi.org/10.1001/jama.2010.1210>.
- [12] J. Ryu, B.-H. Lee, D.-H. Kim, sEMG Signal-based lower Limb Human Motion Detection using a top and Slope Feature Extraction Algorithm, *IEEE Signal Processing Letters* 24 (7) (2017) 929–932, <https://doi.org/10.1109/lsp.2016.2636320>.
- [13] H. Zhou, D. Yang, Z. Li, D. Zhou, J. Gao, J. Guan, Locomotion Mode Recognition for walking on three Terrains based on sEMG of lower Limb and Back Muscles, *Sensors* 21 (9) (2021) 2933, <https://doi.org/10.3390/s21092933>.
- [14] Y. Zhao, J. Wang, Y. Zhang, H. Liu, Z. Chen, Y. Lu, Y. Dai, L. Xu, S. Gao, Flexible and Wearable EMG and PSD Sensors Enabled Locomotion Mode Recognition for IoT-Based In-Home Rehabilitation, *IEEE Sensors Journal* 21 (23) (2021) 26311–26319, <https://doi.org/10.1109/jsen.2021.3058429>.
- [15] Y. Chang, L. Wang, M. Li, M. Liu, L. Lin, B. Cui, Q. Liu, Research on gait recognition of surface EMG signal based on MPSO-LSTM algorithm, *Journal of Mechanics in Medicine and Biology* 23 (08) (2023), <https://doi.org/10.1142/s0219519423400663>.
- [16] J. Wang, Y. Dai, X. Si, Analysis and Recognition of Human lower Limb Motions based on Electromyography (EMG) Signals, *Electronics* 10 (20) (2021) 2473, <https://doi.org/10.3390/electronics10202473>.
- [17] J. Wang, Y. Dai, X. Si, Classification and regression of muscle neural signals on human lower extremities via BP_AdaBoost, *Applied Sciences* 12 (12) (2022) 5830, <https://doi.org/10.3390/app12125830>.
- [18] J. Ryu, B.-H. Lee, J. Maeng, D.-H. Kim, sEMG-signal and IMU sensor-based gait sub-phase detection and prediction using a user-adaptive classifier, *Medical Engineering & Physics* 69 (2019) 50–57, <https://doi.org/10.1016/j.medengphy.2019.05.006>.
- [19] L. Meng, J. Pang, Z. Wang, R. Xu, D. Ming, The Role of Surface Electromyography in Data Fusion with Inertial Sensors to Enhance Locomotion Recognition and Prediction, *Sensors* 21 (18) (2021) 6291, <https://doi.org/10.3390/s21186291>.
- [20] F. Duan, Y. Lv, Z. Sun, J. Li, Multi-scale Learning for Multimodal Neurophysiological Signals: Gait Pattern Classification as an example, *Neural Processing Letters* 54 (3) (2022) 2455–2470, <https://doi.org/10.1007/s11063-021-10738-w>.
- [21] H. Zhou, P. Li, S. Zhou, Lower limb motion recognition based on EEG-EMG signal fusion, *Journal of South-Central University for Nationalities (natural Science Edition)* 44 (1) (2025) 58–67, <https://doi.org/10.20056/j.cnki.ZNMZDK.20250108>.
- [22] Y. Lu, H. Wang, Z. Lu, J. Niu, C. Liu, Gait pattern recognition based on electroencephalogram signals with common spatial pattern and graph attention networks, *Engineering Applications of Artificial Intelligence* 141 (2025) 109680, <https://doi.org/10.1016/j.engappai.2024.109680>.
- [23] M. Zhu, X. Guan, Z. Li, L. He, Z. Wang, K. Cai, sEMG-based lower Limb Motion Prediction using CNN-LSTM with improved PCA Optimization Algorithm, *Journal of Bionic Engineering* 2016–01-01(2022–11–11) (2022), <https://doi.org/10.1007/s42235-022-00280-3>.
- [24] F. Wang, J. Lu, Z. Fan, C. Ren, X. Geng, Continuous motion estimation of lower limbs based on deep belief networks and random forest, *Review of Scientific Instruments* 93 (4) (2022), <https://doi.org/10.1063/5.0057478>.
- [25] Z. Lu, S. Chen, J. Yang, C. Liu, H. Zhao, Prediction of lower limb joint angles from surface electromyography using XGBoost, *Expert Systems with Applications* 264 (2025) 125930, <https://doi.org/10.1016/j.eswa.2024.125930>.
- [26] J. Yang, Z. Lu, S. Chen, C. Liu, H. Zhao, Continuous knee joint angle prediction with surface EMG, *Biomedical Signal Processing and Control* 95 (2024) 106354, <https://doi.org/10.1016/j.bspc.2024.106354>.
- [27] Y. Narayan, Comparative analysis of SVM and Naive Bayes classifier for the SEMG signal classification, *Materials Today: Proceedings* 37 (2) (2021) 3241–3245, <https://doi.org/10.1016/j.matpr.2020.09.093>.
- [28] L. Ling, L. Wei, B. Feng, Z. Yu, L. Wang, sEMG-based knee Angle Prediction: an Efficient Framework with XGBoost Feature selection and Multiattention LSTM, *IEEE Sensors Journal* 25 (9) (2025) 16501–16514, <https://doi.org/10.1109/jsen.2025.3553533>.
- [29] J. Wang, L. Wang, S.M. Miran, X. Xi, A. Xue, Prediction of Joint Angles based on Human lower Limb Surface Electromyography, *Sensors* 23 (12) (2023) 5404, <https://doi.org/10.3390/s23125404>.
- [30] L. Tu, B. Fan, M. Du, G. Bao, B. Lv, W. Mao, T. Liu, S. Cai, Prediction of Continuous Joint Angles of the Lower Limb Based on sEMG by Using the ISSA-HKELM Algorithm, *IEEE Sensors Journal* 24 (20) (2024) 32596–32606, <https://doi.org/10.1109/jsen.2024.3446868>.
- [31] H. Zhao, Z. Qiu, D. Peng, F. Wang, Z. Wang, S. Qiu, X. Shi, Q. Chu, Prediction of Joint Angles based on Human lower Limb Surface Electromyography, *Sensors* 23 (12) (2023) 5404, <https://doi.org/10.3390/s23125404>.
- [32] Z. Li, X. Guan, K. Zou, C. Xu, Estimation of knee Movement from Surface EMG using Random Forest with principal Component Analysis, *Electronics* 9 (1) (2019) 43, <https://doi.org/10.3390/electronics9010043>.
- [33] C. Wang, W. Guo, H. Zhang, L. Guo, C. Huang, C. Lin, sEMG-based continuous estimation of grasp movements by long-short term memory network, *Biomedical Signal Processing and Control* 59 (2020) 101774, <https://doi.org/10.1016/j.bspc.2019.101774>.
- [34] C. Ma, C. Lin, O.W. Samuel, L. Xu, G. Li, Continuous estimation of upper limb joint angle from sEMG signals based on SCA-LSTM deep learning approach, *Biomedical Signal Processing and Control* 61 (2020) 102024, <https://doi.org/10.1016/j.bspc.2020.102024>.
- [35] G. Tang, J. Sheng, D. Wang, S. Men, Continuous Estimation of Human Upper Limb Joint Angles by using PSO-LSTM Model, *IEEE Access* 9 (2021) 17986–17997, <https://doi.org/10.1109/access.2020.3047828>.
- [36] Y. Lu, H. Wang, B. Zhou, C. Wei, S. Xu, Continuous and simultaneous estimation of lower limb multi-joint angles from sEMG signals based on stacked convolutional and LSTM models, *Expert Systems with Applications* 203 (2022) 117340, <https://doi.org/10.1016/j.eswa.2022.117340>.
- [37] J. Han, G. Tian, H. Wang, L. Peyrodie, Continuous limb joint angle prediction from sEMG using SA-FAWT and Conv-BiLSTM, *Biomedical Signal Processing and Control* 97 (2020) 106681, <https://doi.org/10.1016/j.bspc.2024.106681>.
- [38] H. Xie, G. Li, X. Zhao, F. Li, Prediction of Limb Joint Angles based on Multi-Source Signals by GS-GRNN for Exoskeleton Wearer, *Sensors* 20 (4) (2020) 1104, <https://doi.org/10.3390/s20041104>.
- [39] Q. Song, X. Ma, Y. Liu, Continuous Online Prediction of Lower Limb Joints Angles Based on sEMG Signals by Deep Learning Approach, 2023 (163) (2023) 107124, <https://doi.org/10.1016/j.combiomed.2023.107124>.
- [40] H.J. Hermens, B. Freriks, C. Disselhorst-Klug, G. Rau, Development of recommendations for SEMG sensors and sensor placement procedures, *Journal of Electromyography and Kinesiology* 10 (5) (2000) 361–374, [https://doi.org/10.1016/s1050-6411\(00\)00027-4](https://doi.org/10.1016/s1050-6411(00)00027-4).
- [41] E. Bernardes, S. Viollet, Quaternion to Euler angles conversion: a direct, general and computationally efficient method, *PLOS ONE* 17 (11) (2022) e0276302–e, <https://doi.org/10.1371/journal.pone.0276302>.

- [42] S. Madgwick, (2010). An efficient orientation filter for inertial and inertial/magnetic sensor arrays.
- [43] Y. Yu, X. Si, C. Hu, J. Zhang, A Review of Recurrent Neural Networks: LSTM Cells and Network Architectures, *Neural Computation* 31 (7) (2019) 1235–1270, https://doi.org/10.1162/neco_a_01199.
- [44] R. Sowmya, M. Premkumar, P. Jangir, Newton-Raphson-based optimizer: a new population-based metaheuristic algorithm for continuous optimization problems, *Engineering Applications of Artificial Intelligence* 128 (2024) 107532, <https://doi.org/10.1016/j.engappai.2023.107532>.
- [45] A. Vabalas, E. Gowen, E. Poliakoff, A.J. Casson, Machine learning algorithm validation with a limited sample size, *PLOS ONE* 14 (11) (2019) e0224365, <https://doi.org/10.1371/journal.pone.0224365>.
- [46] P. Qin, X. Shi, C. Zhang, K. Han, Continuous estimation of the lower-limb multi-joint angles based on muscle synergy theory and state-space model, *IEEE Sensors Journal* 23 (8) (2023) 8491–8503, <https://doi.org/10.1109/jsen.2023.3240170>.
- [47] S. Zong, W. Li, D. Sun, X. Wei, J. Chen, Z. Yue, D. Sun, A study on lower limb movement intention recognition based on multi-source information fusion, *IEEE Access* 13 (2024) 5032–5041, <https://doi.org/10.1109/access.2024.3521510>.
- [48] L. Yang, Z. Shi, R. Jia, J. Kou, M. Du, C. Bian, J. Wang, Multi-branch deep learning neural network prediction model for the development of angular biosensors based on sEMG, *Frontiers in Bioengineering and Biotechnology* 12 (2024) 1492232, <https://doi.org/10.3389/fbioe.2024.1492232>.
- [49] Z. Guo, H. Zheng, H. Wu, J. Zhang, G. Zhou, J. Long, Transferable multi-modal fusion in knee angles and gait phases for their continuous prediction, *Journal of Neural Engineering* 20 (3) (2023) 036019, <https://doi.org/10.1088/1741-2552/accd22>.
- [50] A. Gautam, M. Panwar, D. Biswas, A. Acharyya, MyoNet: A Transfer-Learning-Based LRCN for Lower Limb Movement Recognition and Knee Joint Angle Prediction for Remote Monitoring of Rehabilitation Progress From sEMG, *IEEE Journal of Translational Engineering in Health and Medicine* 8 (2020) 1–10, <https://doi.org/10.1109/jtehm.2020.2972523>.
- [51] L. Jia, Q. Ai, W. Meng, Q. Liu, S.Q. Xie, Individualized Gait Trajectory Prediction based on Fusion LSTM Networks for Robotic Rehabilitation Training, *IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* 2021 (2021) 988–993, <https://doi.org/10.1109/aim46487.2021.9517616>.
- [52] G. Liu, L. Zhang, B. Han, T. Zhang, Z. Wang, P. Wei, sEMG-based Continuous Estimation of knee Joint Angle using Deep Learning with Convolutional Neural Network, in: *In Proceedings of the 2019 IEEE 15th International Conference on Automation Science and Engineering (CASE)*, 2019, pp. 140–145, <https://doi.org/10.1109/coase.2019.8843168>.
- [53] Y. Yang, Q. Tao, S. Li, S. Fan, Emg-based dual-branch deep learning framework with transfer learning for lower limb motion classification and joint angle estimation, *Concurrency and Computation: Practice and Experience* 37 (23–24) (2025) e70263, <https://doi.org/10.1002/cpe.70263>.
- [54] G.J. Rani, M.F. Hashmi, Deep Learning in Electromyography Signal-based lower Limb Angle Prediction and activity Classification, *J Bionic Eng* 23 (2026) 274–290, <https://doi.org/10.1007/s42235-025-00813-6>.
- [55] J. Liang, Z. Shi, F. Zhu, W. Chen, X. Chen, Y. Li, Gaussian process autoregression for joint angle prediction based on sEMG signals, *Frontiers in Public Health* 9 (2021) 567–581, <https://doi.org/10.3389/fpubh.2021.685596>.
- [56] J.A. Brantley, T.P. Luu, S. Nakagome, & J.L. Contreras-Vidal, (2017, October 5–8). Prediction of lower-limb joint kinematics from surface EMG during overground locomotion. In *Proceedings of the 2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Banff, AB, Canada, 5–8 October 2017; Pp. 1705–1709. Doi:10.1109/smc.2017.8122861.
- [57] J. Chen, X. Zhang, Y. Cheng, N. Xi, Surface EMG based continuous estimation of human lower limb joint angles by using deep belief networks, *Biomedical Signal Processing and Control* 40 (2018) 335–342, <https://doi.org/10.1016/j.bspc.2017.10.002>.
- [58] C. Lv, Y. Liu, Y. Chen, L. Xie, SEMG-based lower limb joint angle prediction with muscle synergy spatio-temporal fusion and dwcit model, *Sensors Journal*, IEEE 25 (14) (2025) 27155–27165, <https://doi.org/10.1109/JSEN.2025.3576764>.
- [59] J. Huang, (2023). Research on lower limb joint angle prediction based on surface electromyography signals (Master's thesis, University of Electronic Science and Technology of China). Doi: 10.27005/d.cnki.gdzku.2023.003688.
- [60] N.A.E. Yaakoubi, C. McDonald, O. Lennon, Prediction of Gait Kinematics and Kinetics: A Systematic Review of EMG and EEG Signal Use and Their Contribution to Prediction Accuracy, *Bioengineering* 10 (10) (2023) 1162, <https://doi.org/10.3390/bioengineering10101162>.
- [61] R. Mobarak, A. Tigrini, F. Verdini, A.H. Al-Timemy, S. Fioretti, L. Burattini, et al., A minimal and multi-source recording setup for ankle joint kinematics estimation during walking using only proximal information from lower limb, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 32 (2024) 812–821, <https://doi.org/10.1109/tnsre.2024.3364976>.
- [62] H. Xie, Research on joint-angle prediction based on artificial neural network for above-knee amputees, *Sensors (Basel, Switzerland)* 21 (21) (2021) 7199, <https://doi.org/10.3390/s21217199>.